

# EE6222 Machine Vision Topics 1 to 6

## Study Points and Mock Questions

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### 考点总结与中英对照模拟练习

范围 **Scope:** PDF Topic 1 至 Topic 6 (讲义第 4 至 215 页) | **Contents:** 6 个主题的考点与公式总结、67 道选择题、18 道匹配题、全部答案与解析

## 第一部分 考点总结与核心公式

以下每个主题对应讲义的一章，表中列出需要记住的概念、符号含义和公式。

### Topic 1 Image Fundamentals and Human Perception 图像基础与人类视觉感知

PDF p.4-43

| Key point 考点                                          | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Image and image formation</b><br>图像与成像             | <p>An image is a two-dimensional projection of a three-dimensional scene. Light received or reflected from a point is <math>I(\lambda) = \rho(\lambda)L(\lambda)</math>, where <math>\rho(\lambda)</math> is the reflectivity of the object and <math>L(\lambda)</math> is the spectral energy distribution of the light source.</p> $I(\lambda) = \rho(\lambda)L(\lambda)$ <p>图像是三维场景在二维平面上的投影。物体上某点接收或反射的光为 <math>I(\lambda) = \rho(\lambda)L(\lambda)</math>，其中 <math>\rho(\lambda)</math> 为物体反射率，<math>L(\lambda)</math> 为光源的谱能量分布。</p> <p>The visible spectrum is about 350 nm to 780 nm; colour and brightness depend on wavelength and intensity.</p> <p>可见光波长约为 350 nm 至 780 nm；颜色与亮度分别取决于波长和强度。</p>      |
| <b>Luminance and visual sensitivity</b><br>亮度与视觉灵敏度   | <p>The luminance (intensity) received by a sensor is the integral of the light distribution weighted by the relative spectral sensitivity function <math>V(\lambda)</math> of the visual system; <math>V(\lambda)</math> is a bell-shaped curve.</p> $f(x, y) = \int_0^\infty I(x, y, \lambda)V(\lambda) d\lambda$ <p>传感器接收的亮度（强度）是光分布与视觉系统相对光谱灵敏度函数 <math>V(\lambda)</math> 的加权积分；<math>V(\lambda)</math> 为钟形曲线。</p>                                                                                                                                                                                                                                                                                             |
| <b>Human perception and colour image</b><br>人类感知与彩色图像 | <p>The retina has three types of cones (L: red, M: green, S: blue) with different sensitivity functions, so colour is a function of wavelength and three values per sample (pixel) are needed for a colour image.</p> $f_1 = \int_0^\infty I V_1 d\lambda, f_2 = \int_0^\infty I V_2 d\lambda, f_3 = \int_0^\infty I V_3 d\lambda$ <p>视网膜有三种视锥细胞（L 红、M 绿、S 蓝），其灵敏度函数不同，因此颜色是波长的函数，彩色图像每个像素需要三个数值。</p>                                                                                                                                                                                                                                                                                                             |
| <b>Colour space</b><br>颜色空间                           | <p>Additive primaries are R, G, B; subtractive primaries are C, M, Y. A colour is specified as <math>c = ap_1 + bp_2 + cp_3</math>, and a colour space is a three-dimensional space that describes colour in a standard way.</p> $c = ap_1 + bp_2 + cp_3$ <p>加色三原色为 R、G、B；减色三原色为 C、M、Y。颜色可表示为 <math>c = ap_1 + bp_2 + cp_3</math>，颜色空间是标准化描述颜色的三维空间。</p> <p>RGB: hardware oriented (monitors, cameras); rgb: normalized RGB; CMY: colour printers; YIQ: colour TV broadcast; HSI/HSV: colour manipulation; CIE-Luv and CIE-Lab: colour differentiation; sRGB: device-independent display.</p> <p>RGB 面向硬件（显示器、摄像机）；rgb 为归一化 RGB；CMY 用于彩色打印机；YIQ 用于彩色电视广播；HSI/HSV 用于颜色处理；CIE-Luv、CIE-Lab 用于颜色区分；sRGB 用于设备无关显示。</p> |

| Key point 考点                                                | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Digital image representation</b><br>数字图像表示               | <p>A digital image is a sampled, quantized version of a two-dimensional light-intensity function <math>f(x,y)</math>; <math>x</math>, <math>y</math> and <math>f</math> are all finite and discrete. It is usually sampled on an equally spaced rectangular grid, and <math>f(x,y)</math> gives the intensity (brightness, gray level) at that point.</p> <p>数字图像是二维光强函数 <math>f(x,y)</math> 经采样、量化后的结果，<math>x</math>、<math>y</math>、<math>f</math> 都是有限离散量；通常在等间隔矩形网格上采样，<math>f(x,y)</math> 表示该点的强度（亮度、灰度）。</p>                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Digitization: sampling and quantization</b><br>数字化：采样与量化 | <p>Digitization has two parts: spatial sampling (discretization) of the coordinates, and intensity or gray-level quantization of the amplitude.</p> <p>number of gray levels <math>g = 2^b</math>, <math>b</math> bits per sample</p> <p>数字化包括两部分：对空间坐标做采样（离散化），对幅度做强度（灰度）量化。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Spatial and gray-level resolution</b><br>空间分辨率与灰度分辨率     | <p>The spatial resolution refers to the size of the <math>m \times n</math> array on which the image is sampled; the gray-level resolution refers to the number of gray levels <math>g = 2^b</math>, where <math>b</math> is the number of bits per sample.</p> <p><math>g = 2^b</math> (<math>b</math> = bits per sample)</p> <p>空间分辨率指采样图像阵列的大小 <math>m \times n</math>；灰度分辨率指灰度级数 <math>g = 2^b</math>，<math>b</math> 为每样本比特数。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Image histogram</b><br>图像直方图                             | <p>The histogram of an image with gray-level range <math>[0, L]</math> is a discrete function <math>\frac{n_f}{n}</math>, where <math>n_f</math> is the number of pixels with gray level <math>f</math> and <math>n</math> is the total number of pixels; it is an estimate of the probability of occurrence of gray level <math>f</math>.</p> <p><math>p_f(f) = \frac{n_f}{n}</math>, <math>p_f(f) \geq 0</math>, <math>\sum_f p_f(f) = 1</math></p> <p>灰度范围为 <math>[0, L]</math> 的图像直方图是离散函数 <math>\frac{n_f}{n}</math>，<math>n_f</math> 为灰度 <math>f</math> 的像素数，<math>n</math> 为总像素数；它也是灰度 <math>f</math> 出现概率的估计。</p> <p>A narrowly distributed histogram indicates a low-contrast image; a bimodal histogram suggests that the image contains an object with a narrow amplitude range against a background of different amplitude.</p> <p>直方图分布很窄说明图像对比度低；双峰直方图说明图像中存在窄幅值范围的物体以及不同幅值的背景。</p> |
| <b>Image processing and applications</b><br>图像处理及其应用        | <p>Digital image processing is a series of machine or computer operations that produce a modified image or extract more meaningful information (features), or understand (recognize) the image content.</p> <p>数字图像处理是一系列计算机操作，用来产生修改后的图像、提取更有意义的信息（特征），或理解（识别）图像内容。</p> <p>Typical uses: visualization (contrast enhancement, denoising, pseudo colouring), image understanding (colour, shape, texture, edges, lines, curves, corners), automated guided vehicles, visual servicing and robot control, security and biometrics, content-based image retrieval.</p> <p>典型应用：可视化（对比度增强、去噪、伪彩色）、图像理解（颜色、形状、纹理、边缘、线、曲线、角点）、自动导引车、视觉服务与机器人控制、安全与生物特征识别、基于内容的图像检索。</p>                                                                                                                                                                                                                                |

Topic 1 questions in the quiz usually test the meaning of a term or a symbol in a formula ( $\rho$ ,  $L$ ,  $V$ , three cones,  $2^b$ ,  $\frac{n_f}{n}$ ) and the reading of a histogram.

Topic 1 的考题多集中在公式中各符号的含义 (  $\rho$ 、 $L$ 、 $V$ 、三种视锥、 $2^b$ 、 $\frac{n_f}{n}$  ) , 以及直方图形状的解读。

Topic 2 LSI Systems and Transforms 线性平移不变系统与变换

PDF p.44-71

| Key point 考点                                                        | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Impulse and image decomposition</b><br>冲激函数与图像分解                 | <p>The basic element image is the impulse <math>\delta(x,y) = 1</math> at (0,0) and 0 otherwise. Any image can be written as a sum of shifted and scaled impulses.</p> $f(x,y) = \sum_{j=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} f(i,j) \delta(x-i,y-j)$ <p>图像的基本元素是冲激函数 <math>\delta(x,y)</math> : 在原点为 1 , 其他位置为 0。任意图像都可表示为一组平移并缩放后的冲激之和。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>2-D convolution and impulse response</b><br>二维卷积与冲激响应            | <p>For a linear shift-invariant (LSI) system, the output is the input convolved with the impulse response <math>h(x,y)</math>, which is the output image when an impulse image is input. The LSI system is completely characterized by <math>h(x,y)</math>.</p> $g(x,y) = f(x,y) * h(x,y) = \sum_j \sum_i h(i,j) f(x-i,y-j)$ <p>对线性平移不变 ( LSI ) 系统 , 输出等于输入与冲激响应 <math>h(x,y)</math> 的卷积 ; <math>h(x,y)</math> 就是输入冲激图像时的输出。LSI 系统完全由 <math>h(x,y)</math> 决定。</p> <p><math>h(x,y)</math> is also an image, called the spatial representation of a filter, filter mask, filter coefficients or filter parameters; it is usually small (<math>3 \times 3, 5 \times 5, \dots 11 \times 11</math>).</p> <p><math>h(x,y)</math> 本身也是一幅图像 , 称为滤波器的空间表示、滤波掩模、滤波系数或滤波参数 ; 其尺寸通常较小 ( <math>3 \times 3</math>、<math>5 \times 5</math>.....<math>11 \times 11</math> ) 。</p> |
| <b>Convolution properties and numerical evaluation</b><br>卷积性质与数值计算 | <p>Convolution is commutative, associative and distributive. At each point the filter response is the sum of products of the mask coefficients and the corresponding image pixels spanned by the mask.</p> $f * h = h * f, f * (h_1 * h_2) = (f * h_1) * h_2, f * (h_1 + h_2) = f * h_1 + f * h_2$ <p>卷积满足交换律、结合律与分配律。滤波器在每一点的响应是掩模系数与掩模覆盖区域内对应像素的乘积之和。</p> <p>A simple 5-point cross mask averages a pixel with its four neighbours to suppress random noise and smooth the image.</p> $h = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ <p>简单的五点十字掩模把像素与上下左右四个邻居相加 , 用于抑制随机噪声、平滑图像。</p>                                                                                                                                                                                                                                                   |

| Key point 考点                      | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Fourier transform</b><br>傅里叶变换 | <p>The 2-D Fourier transform maps the image to the spatial frequency domain with two frequency variables <math>u</math> and <math>v</math>; the transform is separable, and <math>F(u, v)</math> is in general complex.</p> $F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \exp[-j2\pi(ux + vy)] dx dy$ $f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) \exp[j2\pi(ux + vy)] du dv$ <p>二维傅里叶变换把图像映射到以 <math>u</math>、<math>v</math> 为频率变量的空间频率域；变换是可分离的，<math>F(u, v)</math> 一般为复函数。</p> <p>Magnitude and phase: <math>F = R + jI =  F  \exp(j\phi)</math>, with <math> F  = \sqrt{R^2 + I^2}</math> and <math>\phi = \arctan\left(\frac{I}{R}\right)</math>.</p> <p><math> F(u, v)  = \sqrt{R^2 + I^2}</math>, <math>\phi(u, v) = \arctan\left(\frac{I}{R}\right)</math></p> <p>幅度与相位：<math>F = R + jI =  F  \exp(j\phi)</math>，其中 <math> F  = \sqrt{R^2 + I^2}</math>，<math>\phi = \arctan\left(\frac{I}{R}\right)</math>。</p> |
| <b>2-D DFT</b><br>二维离散傅里叶变换       | <p>For an <math>m \times n</math> image the DFT and its inverse are finite sums; the 2-D DFT is separable and can be computed as row transforms followed by column transforms.</p> $F(u, v) = \sum_{x=0}^{m-1} \sum_{y=0}^{n-1} f(x, y) \exp\left[-j2\pi\left(\frac{ux}{m} + \frac{vy}{n}\right)\right]$ $f(x, y) = \frac{1}{mn} \sum_{u=0}^{m-1} \sum_{v=0}^{n-1} F(u, v) \exp\left[j2\pi\left(\frac{ux}{m} + \frac{vy}{n}\right)\right]$ <p>对 <math>m \times n</math> 图像，DFT 及其逆变换为有限求和；二维 DFT 可分离，可先做行变换再做列变换。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>DFT properties</b><br>DFT 的性质  | <p>Periodicity: <math>F(u, v) = F(u + m, v) = F(u, v + n) = F(u + m, v + n)</math>. For a real image the DFT is conjugate symmetric: <math>F(u, v) = F^*(-u, -v)</math> and <math> F(u, v)  =  F(-u, -v) </math>.</p> <p><math>F(u, v) = F(u + m, v + n)</math>, <math>F(u, v) = F^*(-u, -v)</math></p> <p>周期性：<math>F(u, v) = F(u + m, v) = F(u, v + n) = F(u + m, v + n)</math>。实图像的 DFT 具共轭对称性：<math>F(u, v) = F^*(-u, -v)</math>，<math> F(u, v)  =  F(-u, -v) </math>。</p> <p>Linearity and scaling, convolution theorem, translation and rotation properties. In digital implementation the convolution theorem requires zero padding to avoid wrap-around error.</p> <p><math>f * g \Leftrightarrow FG</math>, <math>fg \Leftrightarrow F * G</math></p> $f(x - x_0, y - y_0) \Leftrightarrow F(u, v) \exp\left[-j2\pi\left(\frac{x_0 u}{m} + \frac{y_0 v}{n}\right)\right]$ <p>线性与尺度变换、卷积定理、平移与旋转性质。数字实现卷积定理时必须补零（zero padding），否则会出现环绕误差。</p>                   |

| Key point 考点                                    | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Image sampling</b><br>图像采样                   | <p>A discrete image is obtained by sampling a continuous image on a rectangular grid. Multiplying the image by the sampling function replicates its spectrum periodically with spacing <math>\left(\frac{1}{\Delta x}, \frac{1}{\Delta y}\right)</math>.</p> $f_d(m, n) = f_c(m\Delta x, n\Delta y)$ $F_d(u, v) = \frac{1}{\Delta x \Delta y} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} F_c\left(u - \frac{m}{\Delta x}, v - \frac{n}{\Delta y}\right)$ <p>离散图像由连续图像在矩形网格上采样得到。图像与采样函数相乘后，其频谱以 <math>\left(\frac{1}{\Delta x}, \frac{1}{\Delta y}\right)</math> 为间隔周期性重复。</p> <p>A function is band-limited if <math>F(u, v) = 0</math> for <math> u  &gt; U_0</math> and <math> v  &gt; V_0</math>; the bandwidths are <math>2U_0</math> and <math>2V_0</math>.</p> <p>若 <math>F(u, v)</math> 在 <math> u  &gt; U_0</math>、<math> v  &gt; V_0</math> 时为零，则函数为带限函数；其带宽为 <math>2U_0</math> 与 <math>2V_0</math>。</p> |
| <b>Sampling theorem and aliasing</b><br>采样定理与混叠 | <p>A band-limited image can be recovered without error if the sampling rates are greater than the Nyquist rates; the recovered image is obtained by low-pass filtering the sampled image.</p> $f_{xs} = \frac{1}{\Delta x} \geq 2U_0, f_{ys} = \frac{1}{\Delta y} \geq 2V_0$ $H(u, v) = \Delta x \Delta y \text{ inside the band, } f_c(x, y) = f_d(x, y) * h(x, y)$ <p>若采样率高于奈奎斯特率，带限图像可无误差恢复；恢复方法是把采样图像通过低通滤波器。</p> <p>Sampling below the Nyquist rate makes the periodic replications overlap, so the original spectrum is irrevocably lost: this is aliasing. It can be avoided or reduced by low-pass filtering (band-limiting) the image before sampling, at the cost of losing high-frequency detail.</p> <p>采样率低于奈奎斯特率时，频谱的周期复制会重叠，原频谱无法恢复，这就是混叠。可在采样前先做低通滤波（限带）来避免或减轻混叠，代价是损失高频细节。</p>                                                                                                                             |

Topic 2 questions are often about what  $h(x, y)$  is, one numerical convolution, one DFT property (periodicity, symmetry, convolution theorem, zero padding) and the Nyquist condition.

Topic 2 的考题常见形式： $h(x, y)$  的含义、一次数值卷积、一条 DFT 性质（周期性、对称性、卷积定理、补零）以及奈奎斯特条件。

### Topic 3 Image Enhancement and Denoising 图像增强与去噪

PDF p.72-121

| Key point 考点 | 内容与公式 Content and formulas |
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| Key point 考点                                                          | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Point processing</b><br>点运算                                        | <p>Many enhancement operations are point and memoryless mappings <math>g = T(f)</math> of the input gray level to the output gray level: power (<math>\gamma</math>) transformation, log transformation, piecewise linear transformation and histogram equalization.</p> <p>power: <math>g = cf^\gamma</math>, log: <math>g = c\log(1+f)</math></p> <p>许多增强操作是把输入灰度映射为输出灰度的点运算（无记忆）<math>g = T(f)</math>：幂次（伽马）变换、对数变换、分段线性变换和直方图均衡。</p> <p>Piecewise linear contrast stretching: different linear mappings on different gray-level intervals.</p> $g = T(f) = \begin{cases} \alpha f, & 0 \leq f < a \\ \beta(f - a) + T(a), & a \leq f < b \\ \gamma(f - b) + T(b), & b \leq f < L \end{cases}$ <p>分段线性对比度拉伸：在不同灰度区间使用不同的线性映射。</p>                                                                                                                              |
| <b>Histogram equalization</b><br>直方图均衡                                | <p>Histogram equalization transforms the input gray level so that the output image has an approximately uniform histogram. The discrete transform is the normalized cumulative histogram; the continuous version is the CDF of <math>f</math>, and then <math>p_g(g) = 1</math>.</p> $c(f) = \sum_{t=0}^f p_f(t) = \sum_{t=0}^f \frac{n_t}{n}$ $g = T(f) = \text{round} \left[ \frac{L(c(f) - c_{\min})}{1 - c_{\min}} \right]$ <p>直方图均衡通过变换输入灰度，使输出图像的直方图近似均匀。离散形式用归一化累积直方图，连续形式是 <math>f</math> 的累积分布函数（CDF），此时 <math>p_g(g) = 1</math>。</p> <p><math>c_{\min}</math> is the smallest value of <math>c(f)</math> and <math>\text{round}[]</math> rounds to an integer. Local histogram equalization applies the same idea in local windows.</p> <p><math>c_{\min}</math> 是 <math>c(f)</math> 的最小值，<math>\text{round}[]</math> 表示四舍五入取整。局部直方图均衡则在局部窗口内做同样处理。</p> |
| <b>Linear filtering: spatial and frequency domain</b><br>线性滤波：空域与频域   | <p>Linear filtering can be done in the spatial domain by convolution or in the frequency domain by multiplication; the two are linked by the convolution theorem.</p> $g(x,y) = f(x,y) * h(x,y) \Leftrightarrow G(u,v) = F(u,v)H(u,v)$ <p>线性滤波可在空域用卷积完成，也可在频域用乘积完成，两者由卷积定理联系。</p> <p>Any linear filter output is a weighted average of the input pixels, so linear filters blur the image, lose sharp details and cannot smooth strong noise.</p> <p>任何线性滤波器的输出都是输入像素的加权平均，因此线性滤波会模糊图像、丢失细节，且难以平滑强噪声。</p>                                                                                                                                                                                                                                                                                                                                                         |
| <b>Smoothing: ideal and Gaussian low-pass filters</b><br>平滑：理想低通与高斯低通 | <p>Smoothing filters (averaging or low-pass filters) are used for blurring and noise reduction. The ideal low-pass filter has a sharp cut-off and produces ringing artifacts; the Gaussian low-pass filter gives a smooth frequency response without ringing.</p> $\text{ideal LP: } H(u,v) = \begin{cases} 1, & D(u,v) \leq D_0 \\ 0, & \text{otherwise} \end{cases}, D(u,v) = \sqrt{u^2 + v^2}$ $\text{Gaussian LP: } G(u,v) = \exp \left[ -\frac{u^2 + v^2}{2D_0^2} \right]$ <p>平滑滤波器（均值或低通滤波器）用于模糊与降噪。理想低通滤波器截止很陡，会产生振铃效应；高斯低通滤波器频率响应平滑，无振铃。</p>                                                                                                                                                                                                                                                                                                              |

| Key point 考点                                                                             | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Sharpening: high-pass and high-boost filtering</b><br>锐化：高通与高频提升                      | <p>A high-pass filter is one minus the corresponding low-pass filter; the Gaussian high-pass filter removes the low frequencies. High-boost filtering adds a scaled copy of the original image back so that low frequencies are retained.</p> $H_{hp}(u,v) = 1 - H_{lp}(u,v)$ <p>Gaussian HP: <math>1 - \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]</math></p> <p>high-boost: <math>f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}, A \geq 1</math></p> <p>高通滤波器等于 1 减去对应的低通滤波器；高斯高通滤波器去除低频。高频提升滤波把原图按比例加回，从而保留低频成分。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Order-statistic filters and the median filter</b><br>排序统计滤波器与中值滤波                     | <p>Order-statistic filters rank the pixels in the filter window. The best-known example is the median filter, which replaces a pixel by the median gray level of its neighbourhood.</p> $\hat{f}(x, y) = \underset{(s,t) \in S_{xy}}{\text{median}} f(s, t)$ <p>排序统计滤波器对滤波窗口内的像素排序。最典型的例子是中值滤波，它用邻域灰度的中值替代该像素。</p> <p>Example: for the nine values (10,15,20,20,20,20,20,25,100) the median is 20, so the value 15 is replaced by 20.</p> <p>例：九个值 (10,15,20,20,20,20,20,25,100) 的中值为 20，因此 15 被替换为 20。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Median filter properties and comparison with the mean filter</b><br>中值滤波的性质及其与均值滤波的比较 | <p>Isolated clusters that are lighter or darker than their neighbours and whose area is less than <math>\frac{n^2}{2}</math> (half the filter area) are eliminated by an <math>n \times n</math> median filter. The median filter preserves step edges, whereas the mean filter blurs edges and pulses.</p> <p>isolated cluster area <math>&lt; \frac{n^2}{2} \rightarrow</math> eliminated by an <math>n \times n</math> median filter</p> <p>面积小于 <math>\frac{n^2}{2}</math>（滤波面积的一半）的孤立亮块或暗块会被 <math>n \times n</math> 中值滤波消除。中值滤波能保持阶跃边缘，而均值滤波会模糊边缘和脉冲。</p> <p>For a step edge window {0,0,0,1,1,1,1} the mean filter output is 0.57 and the median filter output is 1; for {0,0,0,0,1,1,1} they are 0.43 and 0. The median filter is excellent for impulsive (salt-and-pepper) noise but is not as good as the mean filter for Gaussian (short-tailed) noise.</p> <p>对阶跃边缘窗口 {0,0,0,1,1,1,1}，均值滤波输出为 0.57，中值滤波输出为 1；对 {0,0,0,0,1,1,1} 则分别为 0.43 与 0。中值滤波对脉冲（椒盐）噪声效果极好，但对高斯（短尾）噪声不如均值滤波。</p> <p>Repeated applications of a median filter produce a root signal, which is invariant to further filtering; constant and monotonic signals are invariant to the median filter.</p> <p>反复应用中值滤波会得到根信号（root signal），它不再随滤波改变；常数信号与单调信号是中值滤波的不变信号。</p> |



| Key point 考点                                                                      | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other order-statistic and mean-median hybrid filters</b><br>其他排序统计滤波与均值-中值折中滤波 | <p>Max, min and midpoint filters are simple extensions of the median filter; the <math>\alpha</math>-trimmed mean filter removes the <math>d</math> extreme values and averages the remaining <math>mn - d</math> pixels.</p> <p>max filter: <math>\hat{f} = \max f(s \cdot t)</math>; min filter: <math>\hat{f} = \min f(s \cdot t)</math><br/>           midpoint: <math>\hat{f} = 0.5[\max f(s \cdot t) + \min f(s \cdot t)]</math><br/> <math>\alpha</math>-trimmed mean: <math>\hat{f} = \left(\frac{1}{mn-d}\right) \sum f_r(s \cdot t)</math></p> <p>最大值、最小值和中点滤波是中值滤波的简单推广；阿尔法修剪均值滤波去掉 <math>d</math> 个极值后，对剩余 <math>mn - d</math> 个像素取平均。</p>                                                                                                                                                                                                                                                                                                                                                                             |
| <b>The ITM filter and the mean-median relation</b><br>迭代截断均值（ITM）滤波与均值-中值关系       | <p>Three bounds relating the median <math>\phi</math> and the mean <math>\mu</math> of a finite data set: half the distance between the upper and lower mean, one sample standard deviation, and the mean absolute deviation.</p> $\tau_1 = \frac{\mu_h - \mu_l}{2}, \tau_2 = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}, \tau_3 = \frac{1}{n} \sum_{i=1}^n  x_i - \mu $ <p><math> \phi - \mu  \leq \tau_1,  \phi - \mu  \leq \tau_2,  \phi - \mu  \leq \tau_3; \tau_3 \leq \tau_1, \tau_3 \leq \tau_2</math></p> <p>中值 <math>\phi</math> 与均值 <math>\mu</math> 之差的三个上界：上下均值间距的一半、样本标准差、以及平均绝对偏差。<math>\tau_3</math> 是最紧的上界。</p> <p>The iterative truncated arithmetic mean (ITM) filter computes the mean, cuts off samples beyond a threshold, and repeats; the threshold decreases monotonically to zero and the truncated mean approaches the median arbitrarily closely. It avoids data sorting and is faster than computing the median.</p> <p>迭代截断均值（ITM）滤波先求均值，截去超过阈值的样本，再迭代；阈值单调下降到零，截断均值可任意逼近中值。它无需数据排序，比直接计算中值更快。</p> |

Topic 3 questions are usually a numerical median or histogram-equalization calculation, the purpose of a filter (smoothing, sharpening, ringing) or a mean-versus-median comparison.

Topic 3 的考题多为中值或直方图均衡的数值计算、某滤波器的作用（平滑、锐化、振铃）以及均值与中值的比较。

## Topic 4 From Template Matching to Classification and Machine Learning 从模板匹配到分类与机器学习

PDF p.122-151

| Key point 考点                     | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Template matching</b><br>模板匹配 | <p>A system with <math>c</math> gallery objects <math>G_j</math> recognizes an unknown query object <math>F</math> by finding the gallery object with the smallest distance; this is template matching.</p> $G_k = \underset{j}{\operatorname{argmin}} \ F - G_j\ $ <p>系统在 <math>c</math> 个库对象 <math>G_j</math> 中识别未知查询对象 <math>F</math>，方法是找距离最小的库对象，即模板匹配。</p> |

| Key point 考点                                                            | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Euclidean distance between images</b><br>图像间的欧氏距离                    | <p>A <math>p \times q</math> image is converted into an <math>n = pq</math> dimensional column vector so that two images can be compared as two vectors; the square root is omitted for simplicity.</p> $d = \ F - G\ ^2 = \sum_y \sum_x (f(x,y) - g(x,y))^2 = (f - g)^T (f - g)$ <p>把 <math>p \times q</math> 图像转换为 <math>n = pq</math> 维列向量，从而把两幅图像当作两个向量比较；为简便常省略平方根。</p>                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Normalization and correlation coefficient</b><br>归一化与相关系数            | <p>The raw Euclidean distance is huge for two images showing the same pattern with different brightness and contrast, so images are normalized to zero mean and unit variance (unit length) before comparison.</p> $f_c = f - m_f, \quad f_n = \frac{f_c}{\ f_c\ }$ <p>同一图案在亮度与对比度不同时，原始欧氏距离会非常大，因此比较前先把图像归一化为零均值、单位方差（单位长度）。</p> <p>The correlation coefficient <math>\gamma</math> of the two normalized vectors is a similarity measure between -1 and 1, equal to 1 for identical patterns, so it can be used for template matching.</p> $\gamma = f_n^T g_n = \frac{(f - m_f)^T (g - m_g)}{\ f_c\  \ g_c\ }, \quad -1 \leq \gamma \leq 1$ <p>两个归一化向量的相关系数 <math>\gamma</math> 是介于 -1 与 1 之间的相似度，图案相同时 <math>\gamma = 1</math>，可用于模板匹配。</p> |
| <b>From template matching to classification</b><br>从模板匹配到分类             | <p>An object has many variants, so it cannot be represented by a single image: each object is a class and each image is a sample of that class. Recognizing an unknown object becomes classifying it into one of the <math>c</math> classes.</p> <p>同一个对象有多种变化，不能用单幅图像表示：每个对象是一个类，每幅图像是该类的一个样本。识别未知对象就变成把它分到 <math>c</math> 个类之一。</p> <p>The 1st nearest neighbour (NN) classifier compares the query with every training sample; this is expensive for large classes and over-fits the training data (poor generalization).</p> <p>第一最近邻（NN）分类器把查询样本与每个训练样本比较；当类别样本很多时计算量大，且会过拟合训练数据、泛化能力差。</p>                                                                                                                                                |
| <b>Minimum distance classifier and machine learning</b><br>最小距离分类器与机器学习 | <p>Instead of comparing with every sample, the minimum distance classifier compares the query with each class mean vector; the class means are computed from training data, and this is machine learning.</p> $\mu_j = \left( \frac{1}{q_j} \right) \sum_{x_i \in \omega_j} x_i, \quad \omega_k = \underset{j}{\operatorname{argmin}} (x - \mu_j)^T (x - \mu_j)$ <p>最小距离分类器改为把查询样本与各类均值向量比较；类均值由训练数据算出，这一过程就是机器学习。</p>                                                                                                                                                                                                                                                                                                                          |

| Key point 考点                                                | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mahalanobis distance and covariance</b><br>马氏距离与协方差      | <p>Using only the class mean ignores the data spread. Normalizing the squared Euclidean distance by the class variance (in 1-D) or by the inverse covariance matrix (in multi-D) gives the squared Mahalanobis distance; the minimum Mahalanobis distance classifier is optimal if each class obeys a Gaussian PDF.</p> $d_{Eu,j} = \ x - \mu_j\ ^2 = (x - \mu_j)^T (x - \mu_j)$ $d_{Ma,j} = (x - \mu_j)^T \Sigma_j^{-1} (x - \mu_j)$ <p>只用类均值忽略了数据的分布范围。把平方欧氏距离用类方差（一维）或协方差矩阵的逆（多维）加权，得到平方马氏距离；若各类服从高斯分布，最小马氏距离分类器是最优的。</p> <p>The covariance matrix is computed from the training samples of a class; it describes the shape, size and orientation of the data cloud.</p> $\Sigma_j = \left( \frac{1}{q_j} \right) \sum (x_i - \mu_j)(x_i - \mu_j)^T = \left( \frac{1}{q_j} \right) X_j X_j^T$ <p>协方差矩阵由某一类的训练样本计算得到，描述数据云团的形状、大小与方向。</p> |
| <b>Pooled covariance and scatter matrices</b><br>合并协方差与散布矩阵 | <p>When a class has few training samples its covariance matrix is unreliable, so the covariance matrices are averaged over all classes to give the pooled class covariance matrix, also called the within-class scatter matrix. The total scatter matrix decomposes into within-class and between-class scatter.</p> $\Sigma_w = \left( \frac{1}{q} \right) \sum_j q_j \Sigma_j = S_w$ $S_b = \left( \frac{1}{q} \right) \sum_j q_j (\mu_j - \mu)(\mu_j - \mu)^T, S_t = S_w + S_b$ <p>当某类训练样本很少时，其协方差矩阵不可靠，因此把各类协方差矩阵加权平均得到合并类协方差矩阵，也称类内散布矩阵。总散布矩阵可分解为类内与类间散布之和。</p>                                                                                                                                                                                                                                                                        |

Topic 4 questions are typically about why normalization is needed, what the correlation coefficient means, the definition of the Mahalanobis distance and when it is preferred, and the scatter matrix relations.

Topic 4 的考题通常涉及为何需要归一化、相关系数的含义、马氏距离的定义与适用条件，以及散布矩阵之间的关系。

## Topic 5 MAP Decision and Classifiers 最大后验决策与分类器

PDF p.152-185

| Key point 考点 | 内容与公式 Content and formulas |
|--------------|----------------------------|
|--------------|----------------------------|

| Key point 考点                                            | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intuitive decision and MAP rule</b><br>直观决策与 MAP 准则  | <p>With only prior knowledge, the best decision is the maximum probability decision; after observing data <math>x</math> the priors become posterior probabilities <math>p(\omega_i x)</math>, and the rule becomes the maximum a posteriori (MAP) decision.</p> <p>decide <math>\omega_k</math> if <math>p(\omega_k x) &gt; p(\omega_i x), i \neq k; \omega_k = \underset{i}{\operatorname{argmax}} p(\omega_i x)</math></p> <p>只有先验知识时，最优决策是最大概率决策；观测到数据 <math>x</math> 后，先验变成后验概率 <math>p(\omega_i x)</math>，决策规则变为最大后验（MAP）决策。</p> <p>Example: in a school where 70% of students are male, decide male, and the probability of error is 0.3, which is the minimum possible.</p> <p>例：某学院 70% 学生为男性，则判为男性，错误率为 0.3，这是最小可能的错误率。</p>                                                                                                                            |
| <b>Error rate of a decision</b><br>决策的错误率               | <p>Deciding <math>\omega_k</math> gives correct probability <math>p(\omega_k x)</math> and error <math>p(e x) = 1 - p(\omega_k x)</math>; the MAP rule minimizes this error. The performance of the system is the average of <math>p(e x)</math> weighted by <math>p(x)</math>.</p> $p(e x) = 1 - p(\omega_k x) = \sum_{i \neq k} p(\omega_i x)$ $p(e) = \int p(e x)p(x)dx = 1 - \sum_i p(\omega_i) \int_{R_i} p(x \omega_i)dx$ <p>判为 <math>\omega_k</math> 时，正确概率为 <math>p(\omega_k x)</math>，错误率为 <math>p(e x) = 1 - p(\omega_k x)</math>；MAP 准则使该错误最小。系统性能是 <math>p(e x)</math> 按 <math>p(x)</math> 加权的平均。</p> <p>For two classes: <math>p(e) = p(\omega_1) \int_{R_2} p(x \omega_1)dx + p(\omega_2) \int_{R_1} p(x \omega_2)dx</math>.</p> <p>两类问题时：<math>p(e) = p(\omega_1) \int_{R_2} p(x \omega_1)dx + p(\omega_2) \int_{R_1} p(x \omega_2)dx</math>.</p> |
| <b>Bayes rule and the mixture density</b><br>贝叶斯公式与混合密度 | <p>The posterior is not directly measurable, but the prior <math>p(\omega_i)</math> and the class-conditional density <math>p(x \omega_i)</math> are easier to obtain; the mixture density <math>p(x)</math> has the same value for all classes and can therefore be dropped from the decision.</p> $p(\omega_i x) = \frac{p(\omega_i)p(x \omega_i)}{p(x)}, p(x) = \sum_i p(x \omega_i)p(\omega_i)$ <p>后验概率无法直接测得，但先验 <math>p(\omega_i)</math> 与类条件密度 <math>p(x \omega_i)</math> 更容易获得；混合密度 <math>p(x)</math> 对所有类相同，可在决策中略去。</p>                                                                                                                                                                                                                                                                                                                                |
| <b>Discriminant functions</b><br>判别函数                   | <p>The Bayesian (optimal) classifier evaluates one discriminant function per class and assigns <math>x</math> to the class with the largest value; the natural logarithm is used because it is monotonic and simplifies exponential densities.</p> $g_i(x) = \ln p(x \omega_i) + \ln p(\omega_i)$ <p>贝叶斯（最优）分类器为每一类计算一个判别函数，把 <math>x</math> 判给取值最大的类；取自然对数是因为它单调且能简化指数形式的密度函数。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Key point 考点                                                           | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quadratic classifier for Gaussian classes</b><br>高斯类条件密度下的二次分类器     | <p>If each class-conditional density is multivariate Gaussian <math>N(\mu_i, \Sigma_i)</math>, the discriminant function is a quadratic function of <math>x</math> and the decision boundary is quadratic.</p> $p(x \omega_i) = \left( \frac{1}{(2\pi)^{\frac{d}{2}}  \Sigma_i ^{\frac{1}{2}}} \right) \exp \left[ -\left( \frac{1}{2} \right) (x-\mu_i)^T \Sigma_i^{-1} (x-\mu_i) \right]$ $g_i(x) = -\left( \frac{1}{2} \right) (x-\mu_i)^T \Sigma_i^{-1} (x-\mu_i) + \ln p(\omega_i) - \left( \frac{1}{2} \right) \ln  \Sigma_i  - \left( \frac{d}{2} \right) \ln 2\pi$ $g_i(x) = -\left( \frac{1}{2} \right) d_{\Sigma}(x, \mu_i) + b_i, \quad b_i = \ln p(\omega_i) - \left( \frac{1}{2} \right) \ln  \Sigma_i $ <p>若每类的类条件密度为多维高斯 <math>N(\mu_i, \Sigma_i)</math>，判别函数是 <math>x</math> 的二次函数，决策边界为二次曲面。</p> |
| <b>Mahalanobis distance versus Euclidean distance</b><br>马氏距离与欧氏距离     | <p>The Mahalanobis distance is the squared Euclidean distance weighted by the inverse covariance matrix. In 1-D it is the squared difference divided by the variance. All points on an equal-density ellipse have the same Mahalanobis distance to the mean but different Euclidean distances.</p> $d_{\Sigma}(x, \mu_i) = (x-\mu_i)^T \Sigma_i^{-1} (x-\mu_i); \quad d_{Eu}(x, \mu_i) = (x-\mu_i)^T (x-\mu_i)$ $1\text{-D: } d_{\Sigma} = \frac{(x-\mu_i)^2}{\sigma_i^2}, \quad d_{Eu} = (x-\mu_i)^2$ <p>马氏距离是用协方差矩阵的逆加权的平方欧氏距离。一维时等于差值平方除以方差。等密度椭圆上的点到均值有相同的马氏距离，但欧氏距离不同。</p>                                                                                                                                                                                                                                   |
| <b>Linear classifier (equal covariance matrices)</b><br>线性分类器（协方差矩阵相同） | <p>When all classes share the same covariance matrix, the quadratic terms cancel and the discriminant function becomes linear in <math>x</math>, giving a linear classifier whose decision boundary between two classes is a hyperplane.</p> $g_i(x) = w_i^T x + w_{i0}, \quad w_i = \Sigma^{-1} \mu_i, \quad w_{i0} = -\left( \frac{1}{2} \right) \mu_i^T \Sigma^{-1} \mu_i + \ln p(\omega_i)$ $g_i(x) = g_j(x) \Rightarrow (w_i - w_j)^T x + w_{i0} - w_{j0} = 0$ <p>当各类共享同一协方差矩阵时，二次项相互抵消，判别函数变为 <math>x</math> 的线性函数，得到线性分类器，两类之间的决策边界是超平面。</p> <p>This hyperplane is generally not orthogonal to the line joining the two means; the prior probabilities shift it.</p> <p>该超平面一般不与两均值连线正交；先验概率会使它平移。</p>                                                                                                     |

| Key point 考点                                                                                                                      | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Special case <math>\Sigma_i = \sigma^2 I</math> and the minimum distance classifier</b><br>特例 $\Sigma_i = \sigma^2 I$ 与最小距离分类器 | <p>If all features are statistically uncorrelated and have the same variance, the discriminant function simplifies further and the decision boundary is a hyperplane orthogonal to the line linking the two means.</p> $g_i(x) = \mu_i^T x - \left(\frac{1}{2}\right) \mu_i^T \mu_i + \sigma^2 \ln p(\omega_i)$ <p>若所有特征统计不相关且方差相同，判别函数可进一步简化，决策边界是与两均值连线正交的超平面。</p> <p>If, in addition, all prior probabilities are equal, the optimal rule is to assign <math>x</math> to the class with the nearest mean: the minimum distance classifier, which is essentially template matching with class means as prototypes.</p> $g_i(x) = - (x - \mu_i)^T (x - \mu_i) = - \ x - \mu_i\ ^2$ <p>若再假设各类先验概率相同，则最优规则是把 <math>x</math> 判给均值最近的类，即最小距离分类器；这实际上是以类均值作为原型模板的模板匹配。</p> |
| <b>Non-Gaussian data</b><br>非高斯数据                                                                                                 | <p>For non-Gaussian distributions it is very difficult, if not impossible, to derive a theoretically optimal classifier; one solution is to apply a feature transform that converts the distribution into a Gaussian one.</p> <p>对于非高斯分布，很难（甚至不可能）推导理论上最优的分类器；一种解决办法是做特征变换，把分布转换为高斯分布。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Topic 5 questions usually ask about the MAP rule and error probability, the Bayesian discriminant function, and the three Gaussian cases (quadratic, linear, minimum distance).

Topic 5 的考题通常涉及 MAP 准则与错误概率、贝叶斯判别函数，以及高斯假设下的三种情形（二次、线性、最小距离）。

## Topic 6 Statistical Estimation and Machine Learning 统计估计与机器学习

PDF p.186-215

| Key point 考点                                          | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Estimating densities from samples</b><br>由样本估计概率密度 | <p>To design an automatic recognition system we need the probability density function for all values of <math>x</math>, but in practice the PDF is unknown and can only be estimated from training samples <math>D = \{x_i\}</math>. Determining rules or deterministic values of a random variable from samples is statistical estimation or machine learning.</p> <p>设计自动识别系统需要知道所有 <math>x</math> 上的概率密度函数，但实际中 PDF 未知，只能用训练样本 <math>D = \{x_i\}</math> 估计。由样本确定随机变量的规律或确定值，就是统计估计或机器学习。</p> <p>Instead of the posterior <math>p(\omega_k x)</math>, the system learns the prior probability <math>p(\omega_k)</math> and the class-conditional density <math>p(x \omega_k)</math>; the prior is estimated by the relative frequency of the class.</p> $\hat{p}(\omega_k) = \frac{n_k}{n}$ <p>系统不去直接估计后验 <math>p(\omega_k x)</math>，而是学习先验概率 <math>p(\omega_k)</math> 与类条件密度 <math>p(x \omega_k)</math>；先验用类样本出现的频率估计。</p> |

| Key point 考点                                                            | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nonparametric density estimation</b><br>非参数密度估计                      | <p>The probability that a sample falls in a region <math>R</math> of volume <math>V</math> is approximately <math>p(x)V</math>; if <math>k</math> of <math>n</math> samples fall in <math>R</math>, the estimate of the density is <math>\frac{k}{nV}</math>. No assumption about the model is made, so this is a nonparametric approach.</p> $P = \int_R p(t)dt \approx p(x)V, \hat{P} = \frac{k}{n}, \hat{p}(x) = \frac{k}{nV}$ <p>样本落入体积为 <math>V</math> 的区域 <math>R</math> 的概率约为 <math>p(x)V</math>；若 <math>n</math> 个样本中有 <math>k</math> 个落入 <math>R</math>，则密度估计为 <math>\frac{k}{nV}</math>。该方法不对模型形式作假设，因此是非参数方法。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Parzen-window estimation</b><br>Parzen 窗估计                           | <p>The Parzen-window approach fixes the volume of the cell (a <math>d</math>-dimensional hypercube of side <math>h</math>) and counts how many samples fall inside it. Each sample contributes a kernel centred at the sample, so the estimate is a sum of kernels, which interpolates discrete samples into a continuous density.</p> $K(u) = \begin{cases} 1, &  u_j  < \frac{1}{2} \text{ for all } j \text{ (rectangular kernel)} \\ 0, & \text{otherwise} \end{cases}$ $k = \sum_i K\left(\frac{x-x_i}{h}\right), \hat{p}(x) = \left(\frac{1}{nh^d}\right) \sum_i K\left(\frac{x-x_i}{h}\right)$ <p>Parzen 窗方法固定单元体积（边长为 <math>h</math> 的 <math>d</math> 维超立方体），统计落入其中的样本数。每个样本贡献一个以该样本为中心的核函数，因此估计是核之和，把离散样本插值为连续密度。</p> <p>Any function can serve as the kernel provided it is non-negative and integrates to one; the rectangular kernel gives an unsmoothed estimate, whereas a Gaussian kernel gives a smooth one. The width <math>h</math> controls the amount of smoothing, and the probabilistic neural network and RBF network are almost equivalent to the Parzen-window approach.</p> <p><math>K(u) \geq 0, \int K(u)du = 1</math>; Gaussian kernel <math>K(u) = N(0, I)</math></p> <p>只要满足非负且积分为 1，任何函数都可作为核；矩形核给出不平滑的估计，高斯核给出平滑估计。宽度 <math>h</math> 控制平滑程度；概率神经网络与 RBF 神经网络几乎等价于 Parzen 窗方法。</p> |
| <b>k-nearest-neighbour estimation and classifier</b><br>k 近邻估计与 kNN 分类器 | <p>Instead of a fixed volume, the kNN method fixes the number of samples <math>k</math> and grows the volume <math>V(k)</math> until it encloses <math>k</math> samples. The corresponding classifier assigns <math>x</math> to the class that is most frequent among its <math>k</math> nearest neighbours.</p> $\hat{p}(x) = \frac{k}{nV(k)}$ $\hat{p}(\omega_i x) = \frac{k_i}{k}$ <p><math>k</math> 近邻方法不固定体积，而固定样本数 <math>k</math>，扩大体积 <math>V(k)</math> 直到包含 <math>k</math> 个样本；对应的分类器把 <math>x</math> 判给 <math>k</math> 个最近邻中出现最多的类。</p> <p>The 1st-nearest-neighbour classifier (the NN classifier) is the special case <math>k = 1</math>.</p> <p>第一近邻分类器（NN 分类器）是 <math>k = 1</math> 的特例。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



| Key point 考点                                                        | 内容与公式 Content and formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Error rate of the nearest-neighbour classifier</b><br>最近邻分类器的错误率 | <p>When the number of training samples is very large, the error rate <math>P_c</math> of the NN classifier is bounded by the Bayes error <math>P^*</math>: it is never below the Bayes rate and its upper bound depends on the number of classes.</p> $P^* \leq P_c \leq P^* \left( 2 - \frac{cP^*}{c-1} \right); \quad c = 2: P_c \leq 2P^*(1-P^*)$ <p>当训练样本数非常大时，NN 分类器的错误率 <math>P_c</math> 介于贝叶斯错误率 <math>P^*</math> 与其上界之间：不会低于贝叶斯率，上界与类别数有关。</p>                                                                                                                                                                                                                                                                      |
| <b>Parametric method</b><br>参数方法                                    | <p>A density estimated from a few samples may deviate greatly from the true population density. If general knowledge permits us to model the density analytically with unknown parameters, the problem reduces from estimating an unknown function to estimating a few parameters; neural networks and deep learning are also parametric methods.</p> <p>样本很少时，估计出的密度可能与真实密度相差很大。若根据先验知识可以用带未知参数的解析函数建模，问题就从估计未知函数简化为估计少量参数；神经网络与深度学习也是参数方法。</p>                                                                                                                                                                                                                                                                            |
| <b>Maximum likelihood estimation</b><br>最大似然估计                      | <p>Samples are drawn independently (i.i.d.) from <math>p(x \theta)</math>; the likelihood of <math>\theta</math> with respect to D is the product of the densities at the samples. The ML estimate selects the parameter that makes the observed data most probable; because the logarithm is monotonic, the product becomes a sum.</p> $p(D \theta) = \prod_{k=1}^n p(x_k \theta), \quad \hat{\theta} = \operatorname{argmax}_{\theta} \sum_{k=1}^n \ln p(x_k \theta)$ $\text{gradient condition: } \nabla_{\theta} \ln p(D \theta) = \sum_{k=1}^n \nabla_{\theta} \ln p(x_k \theta) = 0$ <p>样本独立同分布 (i.i.d.) 地来自 <math>p(x \theta)</math>；<math>\theta</math> 关于 D 的似然是各样本处密度的乘积。最大似然估计取使观测数据出现概率最大的参数；由于对数单调，乘积变为求和。</p> |
| <b>ML estimates for the Gaussian density</b><br>高斯密度的最大似然估计         | <p>For a Gaussian population with unknown mean and covariance, solving the gradient equations gives the sample mean and the sample covariance using <math>\frac{1}{n}</math> (not <math>\frac{1}{n-1}</math>).</p> $\hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k$ $\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^T$ $1\text{-D: } \hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})^2$ <p>对均值与协方差未知的高斯总体，解梯度方程得到样本均值与样本协方差，其中分母为 <math>\frac{1}{n}</math> (不是 <math>\frac{1}{n-1}</math>)。</p>                                                                                                                                                                                           |

Topic 6 questions usually contrast the Parzen window with the kNN rule, ask which quantity is fixed in each method, or ask for the ML estimate of a Gaussian parameter.

Topic 6 的考题常对比 Parzen 窗与 kNN 方法、问各方法固定的是哪一个量，或求高斯参数的最大似然估计。



## 第二部分 模拟练习题目

以下是 67 道选择题，题型与线上测验一致：题干为英文并附中文译文，选项 A 至 E 同样给出中英对照。答案与解析见第三部分。

### Topic 1 Image Fundamentals and Human Perception 图像基础与人类视觉感知

1. Over which wavelength range does the visible spectrum lie, according to the notes?

中文：按照讲义，可见光的波长范围是多少？

- A. 100 nm to 400 nm | 100 nm 到 400 nm
- B. 350 nm to 780 nm | 350 nm 到 780 nm
- C. 780 nm to 1500 nm | 780 nm 到 1500 nm
- D. 0.35 mm to 0.78 mm | 0.35 mm 到 0.78 mm
- E. 2.5  $\mu\text{m}$  to 25  $\mu\text{m}$  | 2.5  $\mu\text{m}$  到 25  $\mu\text{m}$

2. In the image formation model  $I(\lambda) = \rho(\lambda)L(\lambda)$ , which term describes the object being imaged?

中文：在成像模型  $I(\lambda) = \rho(\lambda)L(\lambda)$  中，哪一项描述被成像的物体？

- A.  $L(\lambda)$ , the spectral energy distribution of the light source |  $L(\lambda)$ ，光源的谱能量分布
- B.  $\lambda$ , the wavelength of the visible spectrum |  $\lambda$ ，可见光的波长
- C.  $I(\lambda)$ , the light received by the sensor |  $I(\lambda)$ ，传感器接收到的光
- D.  $V(\lambda)$ , the spectral sensitivity of the eye |  $V(\lambda)$ ，眼睛的光谱灵敏度
- E.  $\rho(\lambda)$ , the reflectivity of the object |  $\rho(\lambda)$ ，物体的反射率

3. In the expression for the luminance  $f(x, y) = \int_0^\infty I(x, y, \lambda)V(\lambda) d\lambda$ , what is  $V(\lambda)$ ?

中文：在亮度表达式  $f(x, y) = \int_0^\infty I(x, y, \lambda)V(\lambda) d\lambda$  中， $V(\lambda)$  表示什么？

- A. The reflectivity of the object surface | 物体表面的反射率
- B. The relative spectral sensitivity function of the visual system | 视觉系统的相对光谱灵敏度函数
- C. The spectral energy distribution of the light source | 光源的谱能量分布
- D. The gray-level quantization step of the sensor | 传感器的灰度量化步长
- E. The Fourier transform of the illumination | 光照的傅里叶变换

4. How many values per sample (pixel) are required to represent a colour image?

中文：表示一幅彩色图像，每个采样点（像素）需要几个数值？

- A. 1 | 1 个
- B. 2 | 2 个
- C. 3 | 3 个

**D.** 4 | 4 个

**E.** 256 | 256 个

**5.** Which of the following sets are the subtractive primaries?

中文：下列哪一组是减色三原色？

**A.** Cyan, Magenta, Yellow | 青、品红、黄

**B.** Red, Green, Blue | 红、绿、蓝

**C.** Luminance, In-phase, Quadrature | 亮度、同相、正交

**D.** Hue, Saturation, Intensity | 色调、饱和度、强度

**E.** Long, Medium, Short cones | 长、中、短视锥

**6.** Which colour space is used for colour TV broadcasting, according to the notes?

中文：按照讲义，哪一种颜色空间用于彩色电视广播？

**A.** YIQ | YIQ

**B.** RGB | RGB

**C.** CMY | CMY

**D.** HSI | HSI

**E.** CIE-Lab | CIE-Lab

**7.** Digitization of a two-dimensional light-intensity function  $f(x,y)$  consists of which two operations?

中文：二维光强函数  $f(x,y)$  的数字化包含哪两个操作？

**A.** Convolution and correlation | 卷积与相关

**B.** Low-pass filtering and high-pass filtering | 低通滤波与高通滤波

**C.** Forward DFT and inverse DFT | 正 DFT 与逆 DFT

**D.** Spatial sampling (discretization) and gray-level (intensity) quantization | 空间采样（离散化）与灰度（强度）量化

**E.** Thresholding and edge detection | 阈值分割与边缘检测

**8.** If each sample of a digital image is quantized with  $b = 6$  bits, how many gray levels does the image have?

中文：若数字图像每个采样点用  $b = 6$  比特量化，则该图像的灰度级数是多少？

**A.** 6 | 6

**B.** 12 | 12

**C.** 32 | 32

**D.** 64 | 64

**E.** 256 | 256

**9.** The (spatial) resolution of a digital image refers to:

中文：数字图像的（空间）分辨率指：

**A.** The number of gray levels used in the image | 图像使用的灰度级数

**B.** The number of bits per sample | 每样本的比特数

**C.** The size of the filter mask applied to the image | 所加滤波掩模的尺寸

**D.** The number of colour channels of the image | 图像的彩色通道数

**E.** The size of the  $m \times n$  array on which the image is sampled | 图像采样所用的  $m \times n$  阵列大小

**10.** In an image with  $n = 1000$  pixels, a gray level  $f$  occurs  $n_f = 200$  times. What is the value of the normalized histogram  $p_f(f) = \frac{n_f}{n}$  at that gray level?

中文：在共  $n = 1000$  个像素的图像中，灰度  $f$  出现  $n_f = 200$  次。归一化直方图  $p_f(f) = \frac{n_f}{n}$  在该灰度的取值是多少？

**A.** 0.02 | 0.02

**B.** 0.2 | 0.2

**C.** 0.5 | 0.5

**D.** 2.0 | 2.0

**E.** 200 | 200

**11.** A narrowly distributed image histogram most likely indicates:

中文：直方图分布很窄通常说明：

**A.** A high-contrast image with all gray levels used | 使用了全部灰度的高对比度图像

**B.** A low-contrast image | 图像对比度低

**C.** An image with impulsive (salt-and-pepper) noise | 图像含脉冲（椒盐）噪声

**D.** An image containing an object against a background of different amplitude | 图像中含有与背景幅值不同的物体

**E.** An image with a perfectly uniform histogram | 图像直方图完全均匀

## Topic 2 LSI Systems and Transforms 线性平移不变系统与变换

**12.** What is the basic element image used to decompose a digital image in the notes?

中文：讲义中用哪种基本元素图像来分解数字图像？

**A.** The impulse  $\delta(x,y)$ , equal to 1 at  $(0,0)$  and 0 elsewhere | 冲激函数  $\delta(x,y)$ ，在  $(0,0)$  处为 1，其他位置为 0

**B.** The step function  $u(x,y)$  | 阶跃函数  $u(x,y)$

**C.** The Gaussian function  $G(x,y)$  | 高斯函数  $G(x,y)$

**D.** The sine wave  $\sin(2\pi(ux+vy))$  | 正弦波  $\sin(2\pi(ux+vy))$

**E.** The unit box function of size  $3 \times 3$  |  $3 \times 3$  的单位方框函数

**13.** What does the impulse response  $h(x,y)$  represent in a linear shift-invariant system?

中文：在线性平移不变系统中，冲激响应  $h(x,y)$  代表什么？

**A.** The input image itself | 输入图像本身

**B.** The noise characteristics of the system | 系统的噪声特性

**C.** The frequency domain representation of the image | 图像的频域表示

**D.** The discretization step sizes  $\Delta x$  and  $\Delta y$  | 离散化步长  $\Delta x$  与  $\Delta y$

**E.** The output image when the impulse input is applied; also the filter mask | 输入为冲激时的输出图像，也即滤波掩模

**14.** For a linear shift-invariant system, the output image  $g(x,y)$  is obtained by:

中文：对线性平移不变系统，输出图像  $g(x,y)$  由下式得到：

- A.** Multiplying the input image by the impulse response | 输入图像与冲激响应相乘
- B.** Correlating the input image with its own transpose | 输入图像与自身转置相关
- C.** Taking the Fourier transform of the input image only | 仅对输入图像做傅里叶变换
- D.** Convoluting the input image with the impulse response | 输入图像与冲激响应卷积
- E.** Quantizing the input image to fewer gray levels | 把输入图像量化到更少的灰度级

**15.** Which of the following is NOT a property of two-dimensional convolution?

中文：下列哪一项不是二维卷积的性质？

- A.** Commutative:  $f * h = h * f$  | 交换律:  $f * h = h * f$
- B.** Associative:  $(f * h_1) * h_2 = f * (h_1 * h_2)$  | 结合律:  $(f * h_1) * h_2 = f * (h_1 * h_2)$
- C.** Shift variant: the response changes when the input is shifted | 平移可变: 输入平移后响应发生变化
- D.** Distributive:  $f * (h_1 + h_2) = f * h_1 + f * h_2$  | 分配律:  $f * (h_1 + h_2) = f * h_1 + f * h_2$
- E.** Applying two masks in sequence equals applying their convolution once | 先后使用两个掩模等价于只使用它们的卷积掩模一次

**16.** A  $3 \times 3$  image neighbourhood centred at  $(x,y)$  is  $\begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 7 & 9 & 8 \end{bmatrix}$  and the filter mask is the

5-point cross mask  $h = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ . What is the filter output  $g(x,y)$ ?

中文：以  $(x,y)$  为中心的  $3 \times 3$  图像邻域为  $\begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 7 & 9 & 8 \end{bmatrix}$ ，滤波掩模为五点十字掩模  $h =$

$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ 。滤波输出  $g(x,y)$  是多少？

- A.** 5 | 5
- B.** 22 | 22
- C.** 36 | 36
- D.** 45 | 45
- E.** 90 | 90

**17.** The 2-D Fourier transform of an image is separable, meaning that  $F(u,v) = F_1(u)F_2(v)$  holds:

中文：图像的二维傅里叶变换是可分离的，即  $F(u,v) = F_1(u)F_2(v)$  成立的条件是：

- A.** Always, for any image  $f(x,y)$  | 对任意图像  $f(x,y)$  恒成立
- B.** Only if the image is symmetric | 仅当图像对称时
- C.** Only if the image is band-limited | 仅当图像为带限时

**D.** Only if the image has a uniform histogram | 仅当图像直方图均匀时

**E.** Only if the image is separable, i.e.  $f(x,y) = f_1(x)f_2(y)$  | 仅当图像本身可分离, 即  $f(x,y) = f_1(x)f_2(y)$

**18.** A Fourier transform has real part  $R = 3$  and imaginary part  $I = 4$ . What are its magnitude  $|F|$  and phase  $\phi$ ?

中文: 某傅里叶变换的实部  $R = 3$ 、虚部  $I = 4$ , 其幅度  $|F|$  与相位  $\phi$  分别是多少?

**A.**  $|F| = 7, \phi = \arctan\left(\frac{3}{4}\right)$  |  $|F| = 7, \phi = \arctan\left(\frac{3}{4}\right)$

**B.**  $|F| = 5, \phi = \arctan\left(\frac{4}{3}\right)$  |  $|F| = 5, \phi = \arctan\left(\frac{4}{3}\right)$

**C.**  $|F| = 5, \phi = \arctan\left(\frac{3}{4}\right)$  |  $|F| = 5, \phi = \arctan\left(\frac{3}{4}\right)$

**D.**  $|F| = 12, \phi = \arctan\left(\frac{4}{3}\right)$  |  $|F| = 12, \phi = \arctan\left(\frac{4}{3}\right)$

**E.**  $|F| = 1, \phi = 90 \text{ degrees}$  |  $|F| = 1, \phi = 90 \text{ 度}$

**19.** For an  $m \times n$  image, which property of the DFT allows the spectrum to be computed as row transforms followed by column transforms?

中文: 对  $m \times n$  图像, DFT 的哪条性质使其可以先做行变换再做列变换?

**A.** Separability | 可分离性

**B.** Periodicity | 周期性

**C.** Conjugate symmetry | 共轭对称性

**D.** Rotation property | 旋转性质

**E.** Translation property | 平移性质

**20.** Which statement about the convolution theorem in digital image processing is correct?

中文: 关于数字图像处理中的卷积定理, 下列哪一说法正确?

**A.** Convolution in the spatial domain corresponds to multiplication in the frequency domain, and no padding is ever needed | 空域卷积对应频域相乘, 且从不需要补零

**B.** Convolution in the spatial domain corresponds to convolution in the frequency domain | 空域卷积对应频域卷积

**C.** Multiplication in the spatial domain corresponds to multiplication in the frequency domain | 空域相乘对应频域相乘

**D.** Convolution in the spatial domain corresponds to multiplication in the frequency domain, and zero padding should be applied to avoid errors | 空域卷积对应频域相乘, 应补零以避免误差

**E.** The theorem only holds for continuous functions and never for digital images | 该定理只对连续函数成立, 对数字图像不适用

**21.** A continuous image is band-limited with  $U_0 = 20 \text{ cycles/mm}$  and  $V_0 = 20 \text{ cycles/mm}$ . What is the minimum sampling rate along  $x$  that satisfies the sampling theorem?

中文: 连续图像为带限图像,  $U_0 = 20 \text{ cycles/mm}$ 、 $V_0 = 20 \text{ cycles/mm}$ 。满足采样定理的最小  $x$  方向采样率是多少?

**A.** 10 samples/mm | 10 样本/mm

- B. 20 samples/mm | 20 样本/mm
- C. 40 samples/mm | 40 样本/mm
- D. 80 samples/mm | 80 样本/mm
- E. 400 samples/mm | 400 样本/mm

22. What is aliasing, and how can it be avoided?

中文：什么是混叠？如何避免？

- A. It is caused by sampling above the Nyquist rate and is avoided by increasing the resolution | 由高于奈奎斯特率的采样引起，通过提高分辨率避免
- B. It is the overlap of the periodic replications of the spectrum caused by sampling below the Nyquist rate, and it is reduced by low-pass filtering before sampling | 由低于奈奎斯特率的采样导致频谱周期复制重叠，可通过采样前低通滤波减轻
- C. It is the loss of gray-level resolution caused by quantization, and it is avoided by using more bits | 由量化造成的灰度分辨率损失，通过增加比特数避免
- D. It is the ringing artifact of an ideal low-pass filter | 理想低通滤波器的振铃效应
- E. It is the error caused by applying the convolution theorem without padding | 不补零直接使用卷积定理造成的误差

### Topic 3 Image Enhancement and Denoising 图像增强与去噪

23. Which of the following enhancement operations is NOT a point (memoryless) operation?

中文：下列哪种增强操作不属于点运算（无记忆运算）？

- A. Power ( $\gamma$ ) transformation  $g = cf^\gamma$  | 幂次（伽马）变换  $g = cf^\gamma$
- B. Log transformation  $g = c\log(1+f)$  | 对数变换  $g = c\log(1+f)$
- C. Piecewise linear transformation | 分段线性变换
- D. Median filtering of the neighbourhood | 对邻域做中值滤波
- E. Histogram equalization | 直方图均衡

24. The power transformation used for  $\gamma$  correction is written as:

中文：用于伽马校正的幂次变换表达式为：

- A.  $g = cf + b$  |  $g = cf + b$
- B.  $g = c\log(1+f)$  |  $g = c\log(1+f)$
- C.  $g = c\exp(f)$  |  $g = c\exp(f)$
- D.  $g = \frac{c}{1+f}$  |  $g = \frac{c}{1+f}$
- E.  $g = cf^\gamma$  |  $g = cf^\gamma$

25. What is the main effect of the log transformation  $g = c\log(1+f)$ ?

中文：对数变换  $g = c\log(1+f)$  的主要作用是什么？

- A. It expands the dynamic range and compresses the dark gray levels | 扩展动态范围并压缩暗部灰度
- B. It makes the output histogram exactly uniform | 使输出直方图完全均匀

- C. It removes impulsive noise without blurring | 在不模糊的前提下去除脉冲噪声
- D. It compresses the dynamic range and expands the dark gray levels | 压缩动态范围并扩展暗部灰度
- E. It converts a colour image into a gray-level image | 把彩色图像转换为灰度图像

**26.** What is the objective of histogram equalization, and what is the continuous form of the transform?

中文：直方图均衡的目标是什么？其连续形式的变换是怎样的？

- A. To make the output histogram uniform; the transform is the probability density function of the input gray level | 使输出直方图均匀；变换是输入灰度的概率密度函数
- B. To sharpen the image; the transform is the high-pass filter | 锐化图像；变换是高通滤波器
- C. To make the output histogram uniform; the transform is the cumulative distribution function of the input gray level | 使输出直方图均匀；变换是输入灰度的累积分布函数
- D. To remove noise; the transform is the median of the neighbourhood | 去除噪声；变换是邻域中值
- E. To reduce the number of gray levels; the transform is quantization | 减少灰度级数；变换是量化

**27.** A 4-level image ( $L = 4$ , gray levels 0 to 3) has 16 pixels with histogram counts  $n_0 = 3$ ,  $n_1 = 6$ ,  $n_2 = 5$ ,  $n_3 = 2$ . Using  $g = \text{round} \left[ \frac{L(c(f) - c_{\min})}{1 - c_{\min}} \right]$ , what is the equalized gray level of the input level  $f = 2$ ?

中文：一幅 4 级图像 ( $L = 4$ , 灰度 0 至 3) 共 16 个像素，直方图计数为  $n_0 = 3$ 、 $n_1 = 6$ 、

$n_2 = 5$ 、 $n_3 = 2$ 。用  $g = \text{round} \left[ \frac{L(c(f) - c_{\min})}{1 - c_{\min}} \right]$  计算，输入灰度  $f = 2$  均衡后的灰度是多少？

- A. 1 | 1
- B. 2 | 2
- C. 2.5 | 2.5
- D. 3 | 3
- E. 3.5 | 3.5

**28.** What is the main problem of the ideal low-pass filter used for image smoothing?

中文：用于图像平滑的理想低通滤波器主要存在什么问题？

- A. It cannot remove any noise | 无法去除任何噪声
- B. It produces ringing artifacts because of its sharp cut-off | 由于截止过于陡峭而产生振铃效应
- C. It blurs the image more than any other filter | 比其他滤波器更严重地模糊图像
- D. It amplifies the low frequencies too much | 过度放大了低频成分
- E. It reverses the gray-level order of the image | 使图像灰度顺序反转



29. Which of the following is the Gaussian low-pass filter (GLPF) given in the notes?

中文：下列哪一个是讲义给出的高斯低通滤波器（GLPF）？

- A.  $H(u,v) = \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]$  |  $H(u,v) = \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]$
- B.  $H(u,v) = 1$  if  $D(u,v) \leq D_0$ , else 0 |  $H(u,v) = 1$  ( $D(u,v) \leq D_0$ ) , 否则为 0
- C.  $H(u,v) = 1 - \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]$  |  $H(u,v) = 1 - \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]$
- D.  $H(u,v) = Af - f_{lp}$  |  $H(u,v) = Af - f_{lp}$
- E.  $H(u,v) = \frac{1}{1+\left(\frac{D}{D_0}\right)^{2n}}$  |  $H(u,v) = \frac{1}{1+\left(\frac{D}{D_0}\right)^{2n}}$

30. Which expression describes high-boost filtering, and what is the constraint on A?

中文：高频提升滤波的表达式是什么？对 A 有何约束？

- A.  $f_{hb} = Af - f_{lp}$  with  $0 < A < 1$  |  $f_{hb} = Af - f_{lp}$  , 且  $0 < A < 1$
- B.  $f_{hb} = Af - f_{lp}$  with  $A \geq 1$  |  $f_{hb} = Af - f_{lp}$  , 且  $A \geq 1$
- C.  $f_{hb} = f + f_{lp}$  with  $A \geq 1$  |  $f_{hb} = f + f_{lp}$  , 且  $A \geq 1$
- D.  $f_{hb} = A \log(1+f)$  |  $f_{hb} = A \log(1+f)$
- E.  $f_{hb} = A(f_{lp} - f)$  |  $f_{hb} = A(f_{lp} - f)$

31. A  $3 \times 3$  median filter window contains the nine gray levels (10, 15, 20, 20, 20, 20, 20, 25, 100). What value replaces the centre pixel?

中文：3 × 3 中值滤波窗口内的九个灰度为 (10, 15, 20, 20, 20, 20, 20, 25, 100) , 中心像素被替换为多少？

- A. 15 | 15
- B. 20 | 20
- C. 25 | 25
- D. 27.8 | 27.8
- E. 100 | 100

32. At a step edge the filter window contains (0, 0, 0, 1, 1, 1, 1). What are the outputs of the 7-point mean filter and the 7-point median filter?

中文：在阶跃边缘处，滤波窗口为 (0, 0, 0, 1, 1, 1, 1)。7 点均值滤波与 7 点中值滤波的输出分别是多少？

- A. mean = 1, median = 0.57 | 均值 = 1 , 中值 = 0.57
- B. mean = 0.43, median = 0 | 均值 = 0.43 , 中值 = 0
- C. mean = 0.57, median = 0.57 | 均值 = 0.57 , 中值 = 0.57
- D. mean = 0.57, median = 1 | 均值 = 0.57 , 中值 = 1
- E. mean = 0.5, median = 0.5 | 均值 = 0.5 , 中值 = 0.5

33. An isolated bright cluster is eliminated by an  $n \times n$  median filter when its area is smaller than:

中文：对于  $n \times n$  中值滤波，孤立的亮块在面积小于多少时会被消除？

- A.  $\frac{n}{2}$  pixels |  $\frac{n}{2}$  个像素
- B. n pixels | n 个像素

- C.  $\frac{n^2}{2}$  (one half of the filter area) |  $\frac{n^2}{2}$  (滤波面积的一半)
- D.  $n^2$  pixels (the whole filter area) |  $n^2$  个像素 (整个滤波面积)
- E.  $2n^2$  pixels |  $2n^2$  个像素

34. Which statement about the iterative truncated arithmetic mean (ITM) filter is correct?

中文：关于迭代截断均值 (ITM) 滤波器，下列哪一说法正确？

- A. It avoids data sorting, its truncation threshold decreases monotonically to zero, and the truncated mean approaches the median | 无需排序，截断阈值单调下降至零，截断均值逼近中值
- B. It requires sorting the data, so it is slower than the median filter | 需要排序数据，因此比中值滤波慢
- C. The distance between the median and the mean can be arbitrarily large, so no bound exists | 中值与均值的距离可以任意大，因此不存在上界
- D. Its output is always exactly the arithmetic mean of the window | 其输出总是窗口的算术平均
- E. It can only be applied to one-dimensional signals | 只能用于一维信号

#### Topic 4 From Matching to Classification 从模板匹配到分类与机器学习

35. Template matching recognizes an unknown query object F by:

中文：模板匹配识别未知查询对象 F 的方法是：

- A. Finding the gallery object whose distance to F is largest | 找出与 F 距离最大的库对象
- B. Computing the histogram of F only | 只计算 F 的直方图
- C. Finding the gallery object whose distance to F is smallest | 找出与 F 距离最小的库对象
- D. Applying a low-pass filter to F | 对 F 做低通滤波
- E. Sorting the gray levels of F | 对 F 的灰度排序

36. A gray-level image of size  $p \times q$  is converted into a column vector before comparison. What is the dimension n of that vector?

中文：大小为  $p \times q$  的灰度图像在比较前被转换为列向量，该向量的维数 n 是多少？

- A.  $n = p + q$  |  $n = p + q$
- B.  $n = pq$  |  $n = pq$
- C.  $n = p - q$  |  $n = p - q$
- D.  $n = \max(p, q)$  |  $n = \max(p, q)$
- E.  $n = 2$  |  $n = 2$

37. Which expression gives the (squared) Euclidean distance between two image objects represented by the matrices F and G?

中文：由矩阵 F 和 G 表示的两个图像对象之间的 (平方) 欧氏距离是：

- A.  $d = f^T g$  |  $d = f^T g$

- B.  $d = f + g \mid d = f + g$
- C.  $d = (f+g)^T(f+g) \mid d = (f+g)^T(f+g)$
- D.  $d = \frac{f^T f}{g^T g} \mid d = \frac{f^T f}{g^T g}$
- E.  $d = (f-g)^T(f-g) \mid d = (f-g)^T(f-g)$

38. Two images show the same triangle pattern but with very different brightness and contrast, and their raw Euclidean distance is very large. Why is normalization needed?

中文：两幅图像显示相同的三角形图案，但亮度与对比度差异很大，其原始欧氏距离非常大。为什么需要归一化？

- A. Because the distance should measure the pattern similarity, not the absolute intensity values; normalizing to zero mean and unit variance removes the brightness and contrast differences | 因为距离应衡量图案相似性而非绝对强度；归一化到零均值、单位方差可去除亮度与对比度差异
- B. Because the Euclidean distance is not symmetric | 因为欧氏距离不对称
- C. Because the images have different sizes | 因为两幅图像尺寸不同
- D. Because the Euclidean distance requires both images to be binary | 因为欧氏距离要求图像为二值图像
- E. Because the gray levels must be sorted before comparison | 因为比较前必须对灰度排序

39. Which statement about the correlation coefficient  $\gamma = f_n^T g_n$  used for template matching is correct?

中文：关于用于模板匹配的相关系数  $\gamma = f_n^T g_n$ ，下列哪一说法正确？

- A.  $\gamma$  is always positive and can be larger than 1 |  $\gamma$  总为正，且可大于 1
- B.  $\gamma$  is the Euclidean distance between the two images |  $\gamma$  是两幅图像之间的欧氏距离
- C.  $\gamma$  equals zero for identical patterns | 图案相同时  $\gamma = 0$
- D.  $\gamma$  lies between -1 and 1, and equals 1 when the two normalized patterns are identical |  $\gamma$  介于 -1 与 1 之间，两个归一化图案相同时  $\gamma = 1$
- E.  $\gamma$  can only be used for binary images |  $\gamma$  只能用于二值图像

40. In moving from template matching to classification, which statement describes the change of viewpoint?

中文：从模板匹配走向分类时，视角发生了什么变化？

- A. Each image becomes a class and each object becomes a sample | 每幅图像成为一个类，每个对象成为一个样本
- B. Classes and samples are no longer distinguished | 不再区分类与样本
- C. Each object becomes a class and each image becomes a sample of that class | 每个对象成为一个类，每幅图像成为该类的一个样本
- D. Only one sample per class is used, as in template matching | 与模板匹配一样，每类只用一个样本

E. The distance measure is replaced by a histogram | 距离度量被直方图取代

41. What are the main disadvantages of the 1st nearest neighbour (NN) classifier?

中文：第一近邻 (NN) 分类器的主要缺点是什么？

A. It requires the covariance matrix of each class | 需要每一类的协方差矩阵

B. It assumes a Gaussian distribution for each class | 假设每一类服从高斯分布

C. It can only handle two classes | 只能处理两类问题

D. It cannot be used when images are represented as vectors | 当图像表示为向量时无法使用

E. It must compare the query with every training sample, and it over-fits the training data so its generalization is poor | 需要将查询与每个训练样本比较，且会过拟合训练数据、泛化能力差

42. In the minimum Euclidean distance classifier, the class mean vector of class  $\omega_j$  is computed as:

中文：在最小欧氏距离分类器中，类  $\omega_j$  的均值向量计算为：

A.  $\mu_j$  = the largest sample of class  $\omega_j$  |  $\mu_j$  = 类  $\omega_j$  的最大样本

B.  $\mu_j = \left(\frac{1}{q_j}\right) \sum$  all samples  $x_i$  of class  $\omega_j$  |  $\mu_j = \left(\frac{1}{q_j}\right) \sum$  (类  $\omega_j$  的所有样本  $x_i$ )

C.  $\mu_j$  = the covariance matrix of class  $\omega_j$  |  $\mu_j$  = 类  $\omega_j$  的协方差矩阵

D.  $\mu_j$  = the product of all samples of class  $\omega_j$  |  $\mu_j$  = 类  $\omega_j$  所有样本的乘积

E.  $\mu_j$  = the median value of all classes together |  $\mu_j$  = 所有类合并后的中值

43. What is the Mahalanobis distance, as used in the notes, and when is it preferred over Euclidean distance?

中文：按照讲义，马氏距离是什么？何时比欧氏距离更合适？

A. A distance that ignores covariance and only uses the means | 忽略协方差、只用均值的距离

B. A distance weighted by the inverse of the class covariance matrix, preferred when the data have different variances and correlations | 用类协方差矩阵的逆加权的距离，适用于各类方差和相关结构不同的数据

C. A distance computed in the pixel-intensity histogram space | 在像素强度直方图空间中计算的距离

D. A distance based on the number of gradient descent steps | 基于梯度下降步数的距离

E. A distance in the Fourier domain between class templates | 在傅里叶域中计算类模板之间的距离

44. How is the covariance matrix of a class computed from its training samples?

中文：某一类的协方差矩阵如何由其训练样本计算？

A.  $\Sigma_j$  = the mean vector  $\mu_j$  multiplied by itself |  $\Sigma_j$  = 均值向量  $\mu_j$  与自身相乘

- B.  $\Sigma_j$  = the sum of the pixel values of class  $\omega_j$  |  $\Sigma_j$  = 类  $\omega_j$  的像素值之和
- C.  $\Sigma_j$  = the covariance matrix of all classes averaged with equal weight  $\frac{1}{c}$  |  $\Sigma_j$  = 所有类协方差矩阵按  $\frac{1}{c}$  等权平均
- D.  $\Sigma_j$  = the identity matrix for every class |  $\Sigma_j$  = 每一类都取单位矩阵
- E.  $\Sigma_j = \left(\frac{1}{q_j}\right) \sum (x_i - \mu_j)(x_i - \mu_j)^T$  over the  $q_j$  samples of the class |  $\Sigma_j = \left(\frac{1}{q_j}\right) \sum (x_i - \mu_j)(x_i - \mu_j)^T$  , 对类内  $q_j$  个样本求和

45. Which relation between the scatter matrices is correct?

中文：关于散布矩阵之间的关系，下列哪一项正确？

- A. The total scatter matrix is the difference between the within-class and between-class scatter matrices | 总散布矩阵等于类内散布与类间散布之差
- B. The within-class scatter matrix is the product of the total and between-class scatter matrices | 类内散布矩阵等于总散布与类间散布的乘积
- C. The total scatter matrix is the sum of the within-class and between-class scatter matrices | 总散布矩阵等于类内散布与类间散布之和
- D. The between-class scatter matrix is the pooled covariance of all classes | 类间散布矩阵就是所有类的合并协方差
- E. There is no relation between the three scatter matrices | 三者之间没有关系

## Topic 5 MAP Decision and Classifiers 最大后验决策与分类器

46. In a classroom where 70% of the students are male and 30% are female, the best decision for an unclearly seen student is male. What is the probability of error of this decision?

中文：在某课堂中 70% 的学生为男性、30% 为女性，对看不清的学生应判为男性。该决策的错误概率是多少？

- A. 0.0 | 0.0
- B. 0.3 | 0.3
- C. 0.5 | 0.5
- D. 0.7 | 0.7
- E. 1.0 | 1.0

47. The maximum a posteriori (MAP) decision rule is:

中文：最大后验 (MAP) 决策准则是：

- A.  $\omega_k = \underset{i}{\operatorname{argmin}} p(\omega_i | x)$  |  $\omega_k = \underset{i}{\operatorname{argmin}} p(\omega_i | x)$
- B.  $\omega_k = \underset{i}{\operatorname{argmax}} p(\omega_i)$ , ignoring the data  $x$  |  $\omega_k = \underset{i}{\operatorname{argmax}} p(\omega_i)$  , 忽略数据  $x$
- C.  $\omega_k$  = the class with the largest covariance matrix |  $\omega_k$  = 协方差矩阵最大的类
- D.  $\omega_k = \underset{i}{\operatorname{argmax}} p(\omega_i | x) = \underset{i}{\operatorname{argmin}} p(e | x)$  |  $\omega_k = \underset{i}{\operatorname{argmax}} p(\omega_i | x) = \underset{i}{\operatorname{argmin}} p(e | x)$
- E.  $\omega_k$  = the class whose mean is farthest from  $x$  |  $\omega_k$  = 均值离  $x$  最远的类

48. If the decision is  $\omega_k$ , the probability of an error for the observed  $x$  is:

中文：若判决为  $\omega_k$ ，则在观测值  $x$  处的错误概率为：

- A.  $p(e|x) = p(\omega_k|x) \mid p(e|x) = p(\omega_k|x)$
- B.  $p(e|x) = 1 + p(\omega_k|x) \mid p(e|x) = 1 + p(\omega_k|x)$
- C.  $p(e|x) = 1 - p(\omega_k|x) = \sum_{i \neq k} p(\omega_i|x) \mid p(e|x) = 1 - p(\omega_k|x) = \sum_{i \neq k} p(\omega_i|x)$
- D.  $p(e|x) = p(x) \mid p(e|x) = p(x)$
- E.  $p(e|x) = 0$  always  $\mid p(e|x)$  恒为 0

49. The posterior probability  $p(\omega_i|x)$  is computed from the prior and the class-conditional density. Which expression is correct?

中文：后验概率  $p(\omega_i|x)$  由先验与类条件密度计算。下列哪个表达式正确？

- A.  $p(\omega_i|x) = \frac{p(X|\omega_i)}{p(\omega_i)} \mid p(\omega_i|x) = \frac{p(X|\omega_i)}{p(\omega_i)}$
- B.  $p(\omega_i|x) = \frac{p(\omega_i)p(X|\omega_i)}{p(x)}$ , with  $p(x) = \sum_j p(x|\omega_j)p(\omega_j) \mid p(\omega_i|x) = \frac{p(\omega_i)p(X|\omega_i)}{p(x)}$ , 其中  $p(x) = \sum_j p(x|\omega_j)p(\omega_j)$
- C.  $p(\omega_i|x) = p(\omega_i)p(x) \mid p(\omega_i|x) = p(\omega_i)p(x)$
- D.  $p(\omega_i|x) = \frac{p(x)}{p(\omega_i)} \mid p(\omega_i|x) = \frac{p(x)}{p(\omega_i)}$
- E.  $p(\omega_i|x) = p(\omega_i) + p(x|\omega_i) \mid p(\omega_i|x) = p(\omega_i) + p(x|\omega_i)$

50. For a two-class problem, the overall probability of error can be written as:

中文：在两类问题中，总体错误概率可写成：

- A.  $p(e) = p(\omega_1) + p(\omega_2) \mid p(e) = p(\omega_1) + p(\omega_2)$
- B.  $p(e) = p(\omega_1) \int_{R_1} p(x|\omega_1)dx \mid p(e) = p(\omega_1) \int_{R_1} p(x|\omega_1)dx$
- C.  $p(e) = 1 - p(\omega_1) - p(\omega_2) \mid p(e) = 1 - p(\omega_1) - p(\omega_2)$
- D.  $p(e) = p(\omega_1) \int_{R_2} p(x|\omega_1)dx + p(\omega_2) \int_{R_1} p(x|\omega_2)dx \mid p(e) = p(\omega_1) \int_{R_2} p(x|\omega_1)dx + p(\omega_2) \int_{R_1} p(x|\omega_2)dx$
- E.  $p(e) = p(x) \mid p(e) = p(x)$

51. If the feature vector is extended from the height  $x_1$  to the vector  $x = [x_1, x_2]^T$  by adding the hair length, what happens to the classification?

中文：把特征由身高  $x_1$  扩展为向量  $x = [x_1, x_2]^T$ （加入头发长度）后，分类会发生什么变化？

- A. The error probability generally increases because more data are used  $\mid$  由于使用了更多数据，错误概率一般上升
- B. The error probability is unchanged; only the computation becomes longer  $\mid$  错误概率不变，只是计算量增大
- C. The decision rule is no longer the MAP rule  $\mid$  决策规则不再是 MAP 准则
- D. The classes can no longer be separated  $\mid$  两类无法再分开
- E. The error probability generally decreases; the scalar threshold becomes a curve in the plane  $\mid$  错误概率一般会下降；一维的阈值变为平面中的曲线

52. The Bayesian (optimal) classifier evaluates one discriminant function per class. Which definition is used?

中文：贝叶斯（最优）分类器为每一类计算一个判别函数，其定义是：

- A.  $g_i(x) = p(x|\omega_i) - p(\omega_i) \mid g_i(x) = p(x|\omega_i) - p(\omega_i)$
- B.  $g_i(x) = \frac{p(\omega_i)}{p(x|\omega_i)} \mid g_i(x) = \frac{p(\omega_i)}{p(x|\omega_i)}$
- C.  $g_i(x) = \ln p(x|\omega_i) + \ln p(\omega_i) \mid g_i(x) = \ln p(x|\omega_i) + \ln p(\omega_i)$

- D.  $g_i(x) = p(x)\ln p(\omega_i) \mid g_i(x) = p(x)\ln p(\omega_i)$   
 E.  $g_i(x) = p(x|\omega_i) + p(x) \mid g_i(x) = p(x|\omega_i) + p(x)$

53. If each class-conditional density is multivariate Gaussian  $N(\mu_i, \Sigma_i)$ , the discriminant function is:

中文：若各类条件密度为多维高斯  $N(\mu_i, \Sigma_i)$ ，则判别函数是：

- A. A quadratic function of  $x \mid x$  的二次函数  
 B. A linear function of  $x \mid x$  的线性函数  
 C. A constant independent of  $x \mid$  与  $x$  无关的常数  
 D. A logarithmic function of  $x \mid x$  的对数函数  
 E. A function of the number of training samples only  $\mid$  仅是训练样本数的函数

54. In a 1-D problem, class 1 has  $\mu_1 = 70$  and  $\sigma_1^2 = 25$ , class 2 has  $\mu_2 = 100$  and  $\sigma_2^2 = 400$ . Using the squared Mahalanobis distance, how is  $x = 80$  classified?

中文：一维问题中，类 1 的  $\mu_1 = 70$ 、 $\sigma_1^2 = 25$ ，类 2 的  $\mu_2 = 100$ 、 $\sigma_2^2 = 400$ 。用平方马氏距离判断  $x = 80$  属于哪一类？

- A. Class 1, since its squared Mahalanobis distance is 4  $\mid$  类 1，因为平方马氏距离为 4  
 B. Class 1, since its Euclidean distance is the smallest  $\mid$  类 1，因为欧氏距离最小  
 C. Class 2, since its Euclidean distance is the largest  $\mid$  类 2，因为欧氏距离最大  
 D. Cannot be decided because the variances are different  $\mid$  因为两类方差不同，无法判断  
 E. Class 2, since its squared Mahalanobis distance is 1, smaller than 4  $\mid$  类 2，因为平方马氏距离为 1，小于 4

55. When all classes share the same covariance matrix  $\Sigma_i = \Sigma$ , the decision boundary between any two classes is:

中文：当各类共享同一协方差矩阵  $\Sigma_i = \Sigma$  时，任意两类之间的决策边界是：

- A. A hyperplane that is always orthogonal to the line joining the two means  $\mid$  一个总与两均值连线正交的超平面  
 B. A quadratic surface  $\mid$  一个二次曲面  
 C. A sphere centred at the origin  $\mid$  以原点为中心的球面  
 D. A hyperplane, which is generally not orthogonal to the line joining the two means  $\mid$  一个超平面，一般不与两均值连线正交  
 E. There is no decision boundary  $\mid$  不存在决策边界

56. Two classes have equal covariance matrices  $\Sigma = \sigma^2 I$  and equal priors, with means  $\mu_1 = [0.0]^T$  and  $\mu_2 = [4.0]^T$ . Where is the decision boundary?

中文：两类具有相同的协方差矩阵  $\Sigma = \sigma^2 I$  与相同的先验，均值分别为  $\mu_1 = [0.0]^T$ 、 $\mu_2 = [4.0]^T$ 。决策边界在哪里？

- A.  $x_1 = 0 \mid x_1 = 0$

- B.  $x_1 = 2$ , perpendicular to the line joining the two means |  $x_1 = 2$ , 与两均值连线垂直
- C.  $x_1 = 4$  |  $x_1 = 4$
- D.  $x_2 = 0$  |  $x_2 = 0$
- E.  $x_1 + x_2 = 4$  |  $x_1 + x_2 = 4$

## Topic 6 Statistical Estimation and Machine Learning 统计估计与机器学习

57. In the nonparametric estimate  $\hat{p}(x) = \frac{k}{nV}$ ,  $n = 200$  samples are given and a region of volume  $V = 0.5$  contains  $k = 10$  samples. What is the estimated density?

中文：在非参数密度估计  $\hat{p}(x) = \frac{k}{nV}$  中，已知  $n = 200$  个样本，体积  $V = 0.5$  的区域内落入  $k = 10$  个样本，估计的密度是多少？

- A. 0.05 | 0.05
- B. 0.1 | 0.1
- C. 0.2 | 0.2
- D. 0.5 | 0.5
- E. 20 | 20

58. In the Parzen-window approach, the rectangular kernel function of the hypercube of side  $h$  is defined as:

中文：在 Parzen 窗方法中，边长  $h$  的超立方体所用矩形核函数定义为：

- A.  $K(u) = \exp\left(-\frac{u^T u}{2}\right)$  |  $K(u) = \exp\left(-\frac{u^T u}{2}\right)$
- B.  $K(u) = 1$  if  $|u_j| < \frac{1}{2}$  for all  $j$  and 0 otherwise | 当所有  $j$  满足  $|u_j| < \frac{1}{2}$  时  $K(u) = 1$ ，否则为 0
- C.  $K(u) = |u|$  |  $K(u) = |u|$
- D.  $K(u) = \frac{1}{1+u^2}$  |  $K(u) = \frac{1}{1+u^2}$
- E.  $K(u) = u^T u$  |  $K(u) = u^T u$

59. Which conditions must a function satisfy in order to be used as a kernel in the Parzen-window approach?

中文：在 Parzen 窗方法中，函数要作为核函数必须满足什么条件？

- A. It must be symmetric and have zero mean | 必须对称且均值为零
- B. It must be a rectangular function | 必须是矩形函数
- C. It must be non-negative and integrate to one, so that any PDF can serve as a kernel | 必须非负且积分为 1，因此任何概率密度函数都可作为核
- D. It must be differentiable everywhere | 必须处处可微
- E. It must have a compact support of length  $h$  | 其支撑区间长度必须为  $h$

60. What is the essential difference between the Parzen-window approach and the  $k$ -nearest-neighbour approach?

中文：Parzen 窗方法与  $k$  近邻方法最本质的区别是什么？



- A.** Parzen fixes the number of samples and kNN fixes the volume | Parzen 窗固定样本数，而 kNN 固定体积
- B.** Parzen assumes a Gaussian density and kNN does not | Parzen 窗假设高斯密度，而 kNN 不假设
- C.** Parzen requires a fixed global bandwidth and kNN also uses a fixed global bandwidth | Parzen 窗需要固定的全局带宽，kNN 也使用固定的全局带宽
- D.** Parzen can only be used for classification and kNN only for density estimation | Parzen 窗只能用于分类，kNN 只能用于密度估计
- E.** Parzen fixes the cell volume and counts the samples in it, whereas kNN fixes the number of samples and computes the volume that encloses them | Parzen 窗固定单元体积并统计落入的样本数，而 kNN 固定样本数并计算包围这些样本的体积

**61.** In the Parzen-window estimate  $\hat{p}(x) = \left(\frac{1}{nh^d}\right) \sum_i K\left(\frac{x-x_i}{h}\right)$ , what is the role of the width  $h$ ?

中文：在 Parzen 窗估计  $\hat{p}(x) = \left(\frac{1}{nh^d}\right) \sum_i K\left(\frac{x-x_i}{h}\right)$  中，宽度  $h$  起什么作用？

- A.** It is the number of training samples | 它是训练样本数
- B.** It is the dimension of the feature vector | 它是特征向量的维数
- C.** It must always equal  $\frac{1}{2}$  | 它必须恒等于  $\frac{1}{2}$
- D.** It controls the smoothing: a small  $h$  gives a spiky estimate and a large  $h$  gives an over-smoothed estimate | 它控制平滑程度：h 小则估计尖锐，h 大则过度平滑
- E.** It has no effect on the estimate | 它对估计结果没有影响

**62.** Which neural network models are, according to the notes, almost equivalent to the Parzen-window approach?

中文：按讲义，哪类神经网络模型几乎等价于 Parzen 窗方法？

- A.** Probabilistic neural networks and RBF neural networks | 概率神经网络与 RBF 神经网络
- B.** Convolutional neural networks | 卷积神经网络
- C.** Recurrent neural networks | 循环神经网络
- D.** Transformer networks | Transformer 网络
- E.** Hopfield networks | Hopfield 网络

**63.** How does the k-nearest-neighbour classifier assign an unknown sample  $x$ ?

中文：k 近邻分类器如何判定未知样本  $x$  的类别？

- A.** To the class with the largest prior probability only | 仅判给先验概率最大的类
- B.** To the class whose mean is nearest to  $x$  | 判给均值距  $x$  最近的类
- C.** To the class with the smallest covariance matrix | 判给协方差矩阵最小的类
- D.** To the class of the farthest sample | 判给最远样本所属的类
- E.** To the class that is most frequent (majority) among the  $k$  nearest neighbours of  $x$  | 判给  $x$  的  $k$  个最近邻中出现最多的类

**64.** For a very large number of training samples, the error rate  $P_c$  of the nearest neighbour classifier with  $c = 2$  and Bayes error  $P^* = 0.2$  is bounded above by:

中文：当训练样本数非常大时， $c = 2$ 、贝叶斯错误率  $P^* = 0.2$  的最近邻分类器错误率  $P_c$  的上界是多少？

- A. 0.2 | 0.2
- B. 0.24 | 0.24
- C. 0.32 | 0.32
- D. 0.40 | 0.40
- E. 0.50 | 0.50

**65.** In maximum likelihood (ML) estimation, the parameter  $\theta$  is chosen so that:

中文：在最大似然 (ML) 估计中，参数  $\theta$  的选取原则是：

- A. The likelihood of  $\theta$  is minimized | 使  $\theta$  的似然最小
- B. The prior probability of each class is equal | 使各类先验概率相等
- C. The likelihood of  $\theta$  with respect to the training data is maximized | 使  $\theta$  关于训练数据的似然最大
- D. The covariance matrix becomes the identity matrix | 使协方差矩阵变为单位矩阵
- E. The number of parameters is maximized | 使参数个数最多

**66.** What are the ML estimates of the mean and the variance of a univariate Gaussian density?

中文：一维高斯密度的均值与方差的最大似然估计是什么？

- A. mean =  $\left(\frac{1}{n}\right) \sum x_k$  and variance =  $\left(\frac{1}{n-1}\right) \sum (x_k - \text{mean})^2$  | 均值 =  $\left(\frac{1}{n}\right) \sum x_k$ ，方差 =  $\left(\frac{1}{n-1}\right) \sum (x_k - \text{mean})^2$
- B. mean = the median of the samples and variance = 1 | 均值 = 样本中值，方差 = 1
- C. mean = 0 and variance = the maximum sample value | 均值 = 0，方差 = 样本最大值
- D. mean =  $\left(\frac{1}{n}\right) \sum x_k$  and variance =  $\left(\frac{1}{n}\right) \sum (x_k - \text{mean})^2$  | 均值 =  $\left(\frac{1}{n}\right) \sum x_k$ ，方差 =  $\left(\frac{1}{n}\right) \sum (x_k - \text{mean})^2$
- E. mean =  $\left(\frac{1}{n}\right) \sum x_k$  and variance =  $\left(\frac{1}{n}\right) \sum x_k^2$  only | 均值 =  $\left(\frac{1}{n}\right) \sum x_k$ ，方差仅 =  $\left(\frac{1}{n}\right) \sum x_k^2$

**67.** Out of  $n = 200$  training samples,  $n_k = 30$  belong to class  $\omega_k$ . What is the estimated prior probability  $\hat{p}(\omega_k)$ ?

中文：在  $n = 200$  个训练样本中，属于类  $\omega_k$  的有  $n_k = 30$  个，则先验概率的估计  $\hat{p}(\omega_k)$  是多少？

- A. 0.015 | 0.015
- B. 0.15 | 0.15
- C. 0.30 | 0.30
- D. 0.67 | 0.67
- E. 6.67 | 6.67

## 匹配题 Matching questions

先阅读提示 Prompts，再从答案选项 Answers 中为每个提示选出最匹配的一项；配对结果与解析见第三部分。

### 1) Set 1: Image enhancement and denoising filters (Topic 3) 第一组：图像增强与去噪滤波器 (Topic 3)

#### Prompts 提示

P1. Mean filter | 均值滤波

P2. Median filter | 中值滤波

P3. Ideal low-pass filter | 理想低通滤波

P4. Gaussian low-pass filter | 高斯低通滤波

P5. High-boost filter | 高频提升滤波

P6. Iterative truncated mean (ITM) filter | 迭代截断均值 (ITM) 滤波

#### Answers 选项

a) Sharp cut-off in the frequency domain, which produces ringing artifacts | 频域截止陡峭，会产生振铃效应

b) Approaches the median without data sorting, so it is faster than a median filter | 无需排序即可逼近中值，因此比中值滤波更快

c) Attenuates Gaussian noise but blurs edges and spreads impulsive noise | 能抑制高斯噪声，但会模糊边缘并使脉冲噪声扩散

d)  $f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}$  with  $A \geq 1$ , sharpening while keeping low frequencies |  $f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}$ ,  $A \geq 1$ , 在保留低频的同时锐化

e) Removes impulsive (salt-and-pepper) noise and preserves step edges, but is weaker than the mean filter for Gaussian noise | 可去除脉冲 (椒盐) 噪声并保持阶跃边缘，但对高斯噪声不如均值滤波

f) Smooth frequency response  $G = \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]$  used for smoothing | 频率响应平滑  $G = \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]$ , 用于平滑

### 2) Set 2: Distances and classifiers (Topics 4 and 5) 第二组：距离与分类器 (Topic 4 与 Topic 5)

#### Prompts 提示

P1. Squared Euclidean distance | 平方欧氏距离

P2. Squared Mahalanobis distance | 平方马氏距离

P3. Correlation coefficient of normalized images | 归一化图像的相关系数

**P4.** Minimum distance classifier | 最小距离分类器

**P5.** Linear classifier | 线性分类器

**P6.** Quadratic classifier | 二次分类器

**Answers 选项**

**a)** Obtained when all classes share the same covariance matrix; the decision boundary is a hyperplane | 各类共享同一协方差矩阵时得到，决策边界为超平面

**b)**  $d_{Eu} = \|x - \mu\|^2 = (x - \mu)^T (x - \mu)$  |  $d_{Eu} = \|x - \mu\|^2 = (x - \mu)^T (x - \mu)$

**c)** Obtained for Gaussian classes with different covariance matrices; the decision boundary is a quadratic surface | 各类高斯且协方差矩阵不同时得到，决策边界为二次曲面

**d)**  $y = f_n^T g_n$ , between -1 and 1, equal to 1 for identical patterns |  $y = f_n^T g_n$ , 取值在 -1 与 1 之间，图案相同时为 1

**e)**  $d_\Sigma = (x - \mu)^T \Sigma^{-1} (x - \mu)$ , weighted by the inverse covariance |  $d_\Sigma = (x - \mu)^T \Sigma^{-1} (x - \mu)$ , 用协方差矩阵的逆加权

**f)** Assigns  $x$  to the class whose mean is nearest; equivalent to template matching when priors are equal and  $\Sigma = \sigma^2 I$  | 把  $x$  判给均值最近的类；先验相等且  $\Sigma = \sigma^2 I$  时等价于模板匹配

**3) Set 3: Density estimation and estimation theory (Topic 6) 第三组：密度估计与估计理论 (Topic 6)**

**Prompts 提示**

**P1.** Parzen-window estimation | Parzen 窗估计

**P2.** k-nearest-neighbour estimation | k 近邻估计

**P3.** Both Parzen and kNN | Parzen 与 kNN 的共同点

**P4.** Maximum likelihood estimation | 最大似然估计

**P5.** Parametric method | 参数方法

**P6.** Nonparametric method | 非参数方法

**Answers 选项**

**a)** Chooses the parameters that maximize the likelihood of the observed training data | 选择使观测训练数据似然最大的参数

**b)** No assumption on the form of the density; the estimate is determined entirely by the samples | 对密度形式不作假设，估计完全由样本决定

**c)** Density estimated by summing kernels centred at the data points, with a fixed bandwidth  $h$  | 以数据点为中心叠加核函数来估计密度，带宽  $h$  固定

- d)** Assumes an analytical density model with unknown parameters, e.g. a Gaussian with unknown mean and covariance | 假设具有未知参数的解析密度模型，例如均值与协方差未知的高斯分布
- e)** Neither assumes a fixed parametric form for  $p(x)$ , so both are nonparametric approaches | 都不假设  $p(x)$  具有固定参数形式，因此都是非参数方法
- f)** Density estimated with a neighbourhood whose volume  $V(k)$  contains a fixed number  $k$  of samples | 用体积  $V(k)$ 、包含固定数目  $k$  个样本的邻域来估计密度

### 第三部分 答案与解析

#### 选择题答案 Keys for the multiple choice questions

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                            |
|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | B         | <p>The notes define visible light as the spectrum from 350 nm to 780 nm. The visual sensitivity function <math>V(\lambda)</math> is a bell-shaped curve that peaks in the middle of this range.</p> <p>讲义把可见光定义为 350 nm 至 780 nm 的波段，视觉灵敏度函数 <math>V(\lambda)</math> 呈钟形，峰值位于该波段中段。</p>                                                                   |
| 2         | E         | <p><math>I(\lambda) = \rho(\lambda)L(\lambda)</math> separates the object from the illumination: <math>\rho</math> is the reflectivity of the object and <math>L</math> is the spectral energy distribution of the light source.</p> <p><math>I(\lambda) = \rho(\lambda)L(\lambda)</math> 把物体与光源分开：<math>\rho</math> 是物体反射率，<math>L</math> 是光源的谱能量分布。</p> |
| 3         | B         | <p><math>V(\lambda)</math> is the relative spectral sensitivity function of the visual system; it is a bell-shaped curve, so the sensor response is a weighted integral of the light distribution.</p> <p><math>V(\lambda)</math> 是视觉系统的相对光谱灵敏度函数，呈钟形曲线，因此传感器响应是光分布的加权积分。</p>                                                                             |
| 4         | C         | <p>The retina has three types of cones (L, M, S) with different sensitivity functions <math>V_1</math>, <math>V_2</math> and <math>V_3</math>, so three values per pixel are required; a gray-level image needs only one.</p> <p>视网膜有三种灵敏度函数 <math>V_1</math>、<math>V_2</math>、<math>V_3</math> 不同的视锥细胞（L、M、S），因此每个像素需要三个数值；灰度图像只需一个。</p>                 |
| 5         | A         | <p>The additive primaries are R, G and B, while the subtractive primaries are cyan, magenta and yellow (C, M, Y), which are used in colour printers.</p> <p>加色三原色是 R、G、B，减色三原色是青、品红、黄（C、M、Y），用于彩色打印机。</p>                                                                                                                                                 |
| 6         | A         | <p>RGB is hardware oriented, CMY is used by colour printers, YIQ (luminance, in-phase, quadrature) is used for colour TV broadcast, HSI/HSV for colour manipulation, and CIE-Lab/Luv for colour differentiation.</p> <p>RGB 面向硬件，CMY 用于彩色打印机，YIQ（亮度、同相、正交）用于彩色电视广播，HSI/HSV 用于颜色处理，CIE-Lab/Luv 用于颜色区分。</p>                                                 |
| 7         | D         | <p>Digitization of the spatial coordinates is called spatial sampling or discretization, and digitization of the amplitude is called intensity or gray-level quantization.</p> <p>对空间坐标的数字化称为空间采样（离散化），对幅度的数字化称为强度（灰度）量化。</p>                                                                                                                             |
| 8         | D         | <p>The gray-level resolution is <math>g = 2^b</math>, so 6 bits give <math>2^6 = 64</math> gray levels (8 bits would give 256 levels).</p> <p>灰度分辨率为 <math>g = 2^b</math>，因此 6 比特给出 <math>2^6 = 64</math> 个灰度级（8 比特为 256 级）。</p>                                                                                                                          |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                      |
|-----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9         | E         | Spatial resolution is the size of the $m \times n$ sampling array; the number of gray levels $g = 2^b$ is the gray-level resolution, and the two are different concepts.<br>空间分辨率是 $m \times n$ 采样阵列的大小；灰度级数 $g = 2^b$ 属于灰度分辨率，两者是不同概念。                                             |
| 10        | B         | $p_f(f) = \frac{200}{1000} = 0.2$ . The normalized histogram satisfies $p_f(f) \geq 0$ and sums to 1 over all gray levels, and it estimates the probability of gray level f.<br>$p_f(f) = \frac{200}{1000} = 0.2$ 。归一化直方图满足 $p_f(f) \geq 0$ 且对所有灰度求和为 1，它估计灰度 f 出现的概率。              |
| 11        | B         | A narrow histogram means the gray levels occupy only a small range, that is a low-contrast image; a bimodal histogram instead suggests an object with a narrow amplitude range against a background of different amplitude.<br>直方图很窄说明灰度只占据很小范围，即低对比度图像；而双峰直方图说明存在窄幅值范围的物体与不同幅值的背景。 |
| 12        | A         | Any image is the sum of shifted and scaled impulses: $f(x,y) = \sum_j \sum_i f(i,j) \delta(x-i, y-j)$ .<br>任意图像都是平移并缩放后的冲激之和： $f(x,y) = \sum_j \sum_i f(i,j) \delta(x-i, y-j)$ 。                                                                                                    |
| 13        | E         | $h(x,y)$ is defined as the output image for an impulse input; it is also an image, called the filter mask, filter coefficients or filter parameters, and it completely characterizes the LSI system.<br>$h(x,y)$ 定义为输入为冲激时的输出图像；它本身也是一幅图像，称为滤波掩模、滤波系数或滤波参数，并完全决定该 LSI 系统。           |
| 14        | D         | $g(x,y) = f(x,y) * h(x,y) = \sum_j \sum_i h(i,j) f(x-i, y-j)$ ; the response at each point is the sum of products of the mask coefficients and the corresponding pixels.<br>$g(x,y) = f(x,y) * h(x,y) = \sum_j \sum_i h(i,j) f(x-i, y-j)$ ；每一点的响应是掩模系数与对应像素的乘积之和。                   |
| 15        | C         | Convolution is commutative, associative and distributive; shift invariance is a property of the system, not a violation, and is what makes the convolution representation possible.<br>卷积满足交换律、结合律与分配律；平移不变是系统的性质，正是它使卷积表示成立，而不是被卷积破坏的性质。                                           |
| 16        | B         | Only the centre pixel and its four neighbours are used: 4 (up) +1 (left) +3 (centre) +5 (right) +9 (down) = 22. The value 45 would be the sum of all nine pixels.<br>只用到中心像素及其上下左右四个邻居：4 (上) +1 (左) +3 (中心) +5 (右) +9 (下) = 22。45 是九个像素的总和。                                         |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                       |
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| 17        | E         | <p>The integral can always be computed as two successive one-dimensional transforms, but the product form <math>F(u,v) = F_1(u)F_2(v)</math> requires the image itself to be separable.</p> <p>该积分总可以写成两次一维变换，但乘积形式 <math>F(u,v) = F_1(u)F_2(v)</math> 要求图像本身可分离。</p>                                                                                                                                |
| 18        | B         | <p><math> F  = \sqrt{R^2 + I^2} = \sqrt{9 + 16} = 5</math> and <math>\phi = \arctan\left(\frac{I}{R}\right) = \arctan\left(\frac{4}{3}\right)</math>; <math>F = R + jI =  F \exp(j\phi)</math>.</p> <p><math> F  = \sqrt{R^2 + I^2} = \sqrt{9 + 16} = 5</math>, <math>\phi = \arctan\left(\frac{I}{R}\right) = \arctan\left(\frac{4}{3}\right)</math>; <math>F = R + jI =  F \exp(j\phi)</math>.</p> |
| 19        | A         | <p>The 2-D DFT is separable: the exponential factorises into a row term and a column term, so an <math>m \times n</math> transform can be computed as m-point row transforms followed by n-point column transforms.</p> <p>二维 DFT 可分离：指数项分解为行项与列项，因此 <math>m \times n</math> 变换可先做行变换再做列变换。</p>                                                                                                      |
| 20        | D         | <p><math>f * g \Leftrightarrow FG</math> and <math>fg \Leftrightarrow F * G</math>; in digital implementation the DFT is periodic, so zero padding is needed to prevent wrap-around (circular convolution) error.</p> <p><math>f * g \Leftrightarrow FG</math>, <math>fg \Leftrightarrow F * G</math>; 数字实现时 DFT 具有周期性，必须补零以避免环绕（循环卷积）误差。</p>                                                        |
| 21        | C         | <p>The Nyquist rate is twice the bandwidth: <math>f_{xs} = \frac{1}{\Delta x} \geq 2U_0 = 40</math> samples/mm, i.e. the sampling interval must satisfy <math>\Delta x \leq \frac{1}{2U_0} = 0.025</math> mm.</p> <p>奈奎斯特率是带宽的两倍： <math>f_{xs} = \frac{1}{\Delta x} \geq 2U_0 = 40</math> 样本/mm，即采样间隔需满足 <math>\Delta x \leq \frac{1}{2U_0} = 0.025</math> mm。</p>                                   |
| 22        | B         | <p>Sampling below the Nyquist rate makes the periodic replications of <math>F_c(u,v)</math> overlap so that <math>F_c(u,v)</math> is irrevocably lost; low-pass filtering (band-limiting) the image before sampling reduces aliasing at the cost of losing high-frequency detail.</p> <p>采样率低于奈奎斯特率时，<math>F_c(u,v)</math> 的周期复制发生重叠，原频谱无法恢复；采样前先低通滤波（限带）可减轻混叠，代价是损失高频细节。</p>                        |
| 23        | D         | <p>Point operations map each gray level to an output gray level, <math>g = T(f)</math>, using no information from neighbouring pixels; median filtering uses the neighbourhood and is an order-statistic (nonlinear) operation.</p> <p>点运算按 <math>g = T(f)</math> 逐灰度映射，不使用邻域信息；中值滤波使用邻域像素，属于排序统计（非线性）运算。</p>                                                                                        |
| 24        | E         | <p>The power (<math>\gamma</math>) transformation is <math>g = cf^\gamma</math>; changing <math>\gamma</math> brightens or darkens the image, and both the power and the log law are simple point operations.</p> <p>幂次（伽马）变换为 <math>g = cf^\gamma</math>；改变 <math>\gamma</math> 可使图像变亮或变暗。幂次律与对数律都是简单的点运算。</p>                                                                                      |



| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 25        | D         | <p>The logarithm compresses large values and expands small ones, so it compresses the dynamic range and reveals detail in dark regions; it is one of the point operations listed with the power law and histogram equalization.</p> <p>对数压缩大数值、扩展小数值，因此能压缩动态范围并显示暗区细节；它与幂次律、直方图均衡同属讲义中的点运算。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 26        | C         | <p>Histogram equalization aims at a uniform histogram of the output image <math>g(x,y)</math>; the continuous transform <math>g = T(f) = \int_0^f p_f(t)dt</math> is the CDF of <math>f</math>, which gives <math>p_g(g) = 1</math> because <math>\frac{dg}{df} = p_f(f)</math>.</p> <p>直方图均衡的目标是使输出图像 <math>g(x,y)</math> 的直方图均匀；连续形式 <math>g = T(f) = \int_0^f p_f(t)dt</math> 是 <math>f</math> 的累积分布函数，由于 <math>\frac{dg}{df} = p_f(f)</math>，得到 <math>p_g(g) = 1</math>。</p>                                                                                                                                                                                                                                                                                     |
| 27        | D         | <p><math>c(0) = \frac{3}{16}</math>, <math>c(1) = \frac{9}{16}</math>, <math>c(2) = \frac{14}{16}</math>, <math>c_{min} = \frac{3}{16}</math>, so <math>g(2) = \text{round}\left[\frac{3 \times (\frac{11}{16})}{\frac{13}{16}}\right] = \text{round}\left[\frac{33}{13}\right] = \text{round}[2.54] = 3</math>; the mapping is 0→0, 1→1, 2→3, 3→3.</p> <p><math>c(0) = \frac{3}{16}</math>, <math>c(1) = \frac{9}{16}</math>, <math>c(2) = \frac{14}{16}</math>, <math>c_{min} = \frac{3}{16}</math>, 因此 <math>g(2) = \text{round}\left[\frac{3 \times (\frac{11}{16})}{\frac{13}{16}}\right] = \text{round}\left[\frac{33}{13}\right] = \text{round}[2.54] = 3</math>; 映射为 0→0、1→1、2→3、3→3。</p>                                                                    |
| 28        | B         | <p>The ideal low-pass filter <math>H(u,v) = 1</math> for <math>D(u,v) \leq D_0</math> and 0 otherwise has a discontinuous frequency response, which produces ringing (Gibbs) artifacts; the Gaussian low-pass filter avoids this.</p> <p>理想低通滤波器 <math>H(u,v)</math> 在 <math>D(u,v) \leq D_0</math> 时为 1，否则为 0，频率响应不连续，会产生振铃（Gibbs）效应；高斯低通滤波器可避免这一问题。</p>                                                                                                                                                                                                                                                                                                                                                                                                            |
| 29        | A         | <p>The Gaussian low-pass filter is <math>G(u,v) = \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]</math>, equivalently written with <math>\sigma</math> as <math>\frac{1}{2\pi\sigma^2}\exp\left[-\frac{u^2+v^2}{2\sigma^2}\right]</math>; the expression <math>1 - \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]</math> is the Gaussian high-pass filter, and being 1 inside <math>D_0</math> and 0 outside <math>D_0</math> is the ideal low-pass filter.</p> <p>高斯低通滤波为 <math>G(u,v) = \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]</math>，也可写成含 <math>\sigma</math> 的形式 <math>\frac{1}{2\pi\sigma^2}\exp\left[-\frac{u^2+v^2}{2\sigma^2}\right]</math>；<math>1 - \exp\left[-\frac{u^2+v^2}{2D_0^2}\right]</math> 为高斯高通滤波，而 <math>D(u,v) \leq D_0</math> 时为 1、否则为 0 的是理想低通滤波。</p> |
| 30        | B         | <p>High-boost filtering is <math>f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}</math> with <math>A \geq 1</math>, i.e. the sharpened image plus a scaled copy of the original, so that low frequencies are retained.</p> <p>高频提升滤波为 <math>f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}</math>，且 <math>A \geq 1</math>，即在锐化结果上加回一份原图，从而保留低频成分。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 31        | B         | <p>Ranking the nine values puts 20 in the middle, so the median is 20 and the centre pixel is replaced by 20; the mean would be <math>\frac{250}{9} = 27.8</math>, which is pulled up by the outlier 100.</p> <p>九个值排序后中间值为 20，因此中值为 20，中心像素被替换为 20；若取均值则为 <math>\frac{250}{9} = 27.8</math>，会被异常值 100 拉高。</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 32        | D         | <p>The mean is <math>\frac{4}{7} = 0.57</math>, which blurs the edge; the median of the ranked sequence is the fourth value, 1, so the median filter preserves the step edge (for the window (0,0,0,0,1,1,1) the outputs are 0.43 and 0).</p> <p>均值为 <math>\frac{4}{7} = 0.57</math>，会模糊边缘；排序后第四个数为中值 1，因此中值滤波保持阶跃边缘（窗口为 (0,0,0,0,1,1,1) 时分别为 0.43 与 0）。</p>                                                                                                                                                                          |
| 33        | C         | <p>Isolated clusters whose area is less than <math>\frac{n^2}{2}</math>, one half of the filter area, are forced to take the median intensity of the neighbours; larger clusters are affected much less.</p> <p>面积小于 <math>\frac{n^2}{2}</math>（滤波面积的一半）的孤立亮块或暗块会被强制取邻域的中值强度；更大的区域受影响小得多。</p>                                                                                                                                                                                                                                         |
| 34        | A         | <p>Because <math> \phi - \mu </math> is bounded by <math>\tau_1</math>, <math>\tau_2</math> and <math>\tau_3</math> (with <math>\tau_3</math> the tightest bound), the ITM algorithm can truncate the samples iteratively: the threshold decreases monotonically to zero and the truncated mean approaches the median arbitrarily closely, without sorting.</p> <p>由于 <math> \phi - \mu </math> 被 <math>\tau_1</math>、<math>\tau_2</math>、<math>\tau_3</math>（<math>\tau_3</math> 最紧）所界定，ITM 算法可迭代截断样本：阈值单调下降至零，截断均值任意逼近中值，且无需排序。</p> |
| 35        | C         | <p>Template matching compares the query object with each known gallery object <math>G_j</math> and returns the best match: <math>G_k = \underset{j}{\operatorname{argmin}} \ F - G_j\ </math>, which requires a distance or similarity measure.</p> <p>模板匹配把查询对象与每个已知库对象 <math>G_j</math> 比较并返回最佳匹配：<br/> <math>G_k = \underset{j}{\operatorname{argmin}} \ F - G_j\ </math>，因此需要距离或相似度度量。</p>                                                                                                                                        |
| 36        | B         | <p>Each of the <math>p \times q</math> pixels becomes one element of the vector, so <math>n = pq</math>; for example a <math>3 \times 3</math> image becomes a 9-dimensional vector.</p> <p><math>p \times q</math> 个像素各对应向量中的一个元素，因此 <math>n = pq</math>；例如 <math>3 \times 3</math> 图像对应 9 维向量。</p>                                                                                                                                                                                                                                  |
| 37        | E         | <p>The squared Euclidean distance is the sum of squared pixel differences, which in vector form is <math>(f - g)^T(f - g)</math>; the square root is omitted for symbolic simplicity.</p> <p>平方欧氏距离是像素差值的平方和，写成向量形式为 <math>(f - g)^T(f - g)</math>；为简便通常省略平方根。</p>                                                                                                                                                                                                                                                                    |
| 38        | A         | <p>After centralizing (<math>f_c = f - m_f</math>) and normalizing (<math>f_n = \frac{f_c}{\ f_c\ }</math>), the two patterns become identical vectors, <math>d_n = 0</math>, whereas the distance to a genuinely different pattern stays large.</p> <p>先中心化 (<math>f_c = f - m_f</math>) 再归一化 (<math>f_n = \frac{f_c}{\ f_c\ }</math>) 后，两种图案变为相同向量，<math>d_n = 0</math>；与真正不同图案的距离仍然很大。</p>                                                                                                                                         |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 39        | D         | $\gamma = \frac{(f-m_f)^T (g-m_g)}{\ f_c\  \ g_c\ }$ lies between -1 and 1; for the two example images, both with the same triangle pattern, $\gamma = 1$ , so $\gamma$ can serve as a similarity measure.<br>$\gamma = \frac{(f-m_f)^T (g-m_g)}{\ f_c\  \ g_c\ }$ 介于 -1 与 1 之间；对讲义中两幅同为三角形图案的图像， $\gamma = 1$ ，因此可作为相似度度量。                                                                                                                                                           |
| 40        | C         | <p>An object has many variants, so each object is a class and each image of it is a sample; recognizing an unknown query object becomes classifying it into one of the <math>c</math> classes.</p> <p>同一对象有多种变化，因此每个对象是一个类，每幅图像是该类的一个样本；识别未知对象即把它分入 <math>c</math> 个类之一。</p>                                                                                                                                                                                                          |
| 41        | E         | <p>A class may contain thousands or millions of samples, so comparing the query with every training sample is tedious; the NN classifier also over-fits the training data. One solution is the minimum distance classifier that compares the query with each class mean.</p> <p>一个类可能包含成千上万个样本，逐个比较十分繁琐，而且 NN 分类器会过拟合训练数据。一种解决办法是用最小距离分类器，只与各类均值比较。</p>                                                                                                                               |
| 42        | B         | <p>The mean vector is the average of the <math>q_j</math> training samples of class <math>\omega_j</math>, and the classifier decides <math>\omega_k = \underset{j}{\operatorname{argmin}} (x - \mu_j)^T (x - \mu_j)</math>; the mean vectors are parameters learned from the training data.</p> <p>均值向量是类 <math>\omega_j</math> 的 <math>q_j</math> 个训练样本的平均，分类器判定 <math>\omega_k = \underset{j}{\operatorname{argmin}} (x - \mu_j)^T (x - \mu_j)</math>；均值向量是由训练数据学到的参数。</p>         |
| 43        | B         | <p><math>d_{Ma,j} = (x - \mu_j)^T \Sigma_j^{-1} (x - \mu_j)</math> weights the squared Euclidean distance by the inverse covariance matrix; it accounts for the different variances and the correlations between features, and it is the optimal classifier when each class obeys a Gaussian PDF.</p> <p><math>d_{Ma,j} = (x - \mu_j)^T \Sigma_j^{-1} (x - \mu_j)</math> 用协方差矩阵的逆对平方欧氏距离加权，考虑了各特征的不同方差及相互相关性；当各类服从高斯分布时它是最优分类器。</p>                                                   |
| 44        | E         | <p><math>\Sigma_j = \left(\frac{1}{q_j}\right) \sum (x_i - \mu_j)(x_i - \mu_j)^T = \left(\frac{1}{q_j}\right) X_j X_j^T</math>, where <math>X_j</math> holds the centralized training samples of the class as columns; the matrix describes the shape and orientation of the data cloud.</p> <p><math>\Sigma_j = \left(\frac{1}{q_j}\right) \sum (x_i - \mu_j)(x_i - \mu_j)^T = \left(\frac{1}{q_j}\right) X_j X_j^T</math>，其中 <math>X_j</math> 以该类的中心化训练样本为列；该矩阵描述数据云团的形状与方向。</p>    |
| 45        | C         | <p><math>S_t = S_w + S_b</math>, where <math>S_w = \left(\frac{1}{q}\right) \sum_j q_j \Sigma_j</math> is the pooled (within-class) covariance and <math>S_b = \left(\frac{1}{q}\right) \sum_j q_j (\mu_j - \mu)(\mu_j - \mu)^T</math> is the between-class scatter.</p> <p><math>S_t = S_w + S_b</math>，其中 <math>S_w = \left(\frac{1}{q}\right) \sum_j q_j \Sigma_j</math> 为合并（类内）协方差，<math>S_b = \left(\frac{1}{q}\right) \sum_j q_j (\mu_j - \mu)(\mu_j - \mu)^T</math> 为类间散布。</p> |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 46        | B         | Deciding the more probable class gives the minimum error, which is the prior probability of the other class: $p(e) = p(\omega_2) = 0.3$ . The opposite decision would have error 0.7.<br>判为概率更大的类可得到最小错误率，等于另一类的先验概率 $p(e) = p(\omega_2) = 0.3$ ；若作相反判决则错误率为 0.7。                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 47        | D         | The MAP rule chooses the class with the largest posterior probability, which is equivalent to minimizing the probability of a wrong decision; it generalizes the maximum probability decision by replacing the priors with posteriors.<br>MAP 准则选取后验概率最大的类，等价于使错误判决概率最小；它把最大概率决策中的先验替换为后验，从而是其推广。                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 48        | C         | The correct decision probability is $p(\omega_k x)$ , so the error probability is one minus that value, i.e. the sum of the posterior probabilities of all other classes; because $p(\omega_k x)$ is maximal, the error is minimal.<br>正确判决概率为 $p(\omega_k x)$ ，因此错误概率为 1 减去该值，即其他各类后验概率之和；由于 $p(\omega_k x)$ 最大，错误率最小。                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 49        | B         | From $p(x \omega_i) = p(x)p(\omega_i x) = p(\omega_i)p(x \omega_i)$ we obtain the Bayes rule; $p(x)$ is the mixture density, the same for all classes, so it can be removed from the decision.<br>由 $p(x \omega_i) = p(x)p(\omega_i x) = p(\omega_i)p(x \omega_i)$ 得到贝叶斯公式； $p(x)$ 是混合密度，对所有类相同，因此可在判决中约去。                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 50        | D         | The error is the integral of the class-conditional density of class 1 over the decision region assigned to class 2, plus the symmetric term; the correct decision probability is $p(\text{correct}) = 1 - p(e)$ .<br>错误包括：把类 1 的样本落入判为类 2 的区域，以及对称的另一项；正确判决概率为 $p(\text{correct}) = 1 - p(e)$ 。                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 51        | E         | The principle is the same but the scalar $x$ is replaced by the vector $\mathbf{x}$ ; increasing the information (feature dimension) generally increases the probability of a correct decision, and a threshold in 1-D becomes a curve in the plane.<br>识别原理不变，只是把标量 $x$ 换成向量 $\mathbf{x}$ ；增加信息（特征维数）一般会提高正确判决概率，一维的阈值变为平面中的曲线。                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 52        | C         | Because $p(\omega_i x) = p(\omega_i)p(x \omega_i)p(x)^{-1}$ and $p(x)$ is independent of the class, the decision is made by maximising $g_i(x) = \ln p(x \omega_i) + \ln p(\omega_i)$ ; the monotonic logarithm simplifies exponential densities.<br>由于 $p(\omega_i x) = p(\omega_i)p(x \omega_i)p(x)^{-1}$ 且 $p(x)$ 与类别无关，判决等价于最大化 $g_i(x) = \ln p(x \omega_i) + \ln p(\omega_i)$ ；取对数单调且能简化指数形式。                                                                                                                                                                                                                                                                                                                                                                                       |
| 53        | A         | $g_i(x) = -\left(\frac{1}{2}\right)(\mathbf{x}-\boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) + \ln p(\omega_i) - \left(\frac{1}{2}\right) \ln  \boldsymbol{\Sigma}_i  - \left(\frac{d}{2}\right) \ln 2\pi = -\left(\frac{1}{2}\right) d_{\boldsymbol{\Sigma}}(\mathbf{x}, \boldsymbol{\mu}_i) + b_i$ , which is quadratic in $\mathbf{x}$ and gives a quadratic classifier.<br>$g_i(x) = -\left(\frac{1}{2}\right)(\mathbf{x}-\boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) + \ln p(\omega_i) - \left(\frac{1}{2}\right) \ln  \boldsymbol{\Sigma}_i  - \left(\frac{d}{2}\right) \ln 2\pi = -\left(\frac{1}{2}\right) d_{\boldsymbol{\Sigma}}(\mathbf{x}, \boldsymbol{\mu}_i) + b_i$ ，是关于 $\mathbf{x}$ 的二次函数，得到二次分类器。 |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 54        | E         | $d_{Ma1} = \frac{(80-70)^2}{25} = 4 \text{ and } d_{Ma2} = \frac{(80-100)^2}{400} = 1, \text{ so } x \text{ is assigned to class 2}$ <p>even though its Euclidean distance to class 2 (20) is larger than to class 1 (10).</p> $d_{Ma1} = \frac{(80-70)^2}{25} = 4, d_{Ma2} = \frac{(80-100)^2}{400} = 1, \text{ 因此判为类 2——尽管它到类 2 的欧氏距离 20 大于到类 1 的 10。}$                                                                                                                                                                                                                           |
| 55        | D         | <p>With equal covariance matrices the discriminant becomes linear, <math>g_i(x) = w_i^T x + w_{i0}</math> with <math>w_i = \Sigma^{-1} \mu_i</math>, so the boundary <math>(w_i - w_j)^T x + w_{i0} - w_{j0} = 0</math> is a hyperplane; priors shift it so it is generally not orthogonal to the line between the means.</p> <p>协方差相同时判别函数变为线性 <math>g_i(x) = w_i^T x + w_{i0}</math>, 其中 <math>w_i = \Sigma^{-1} \mu_i</math>, 因此边界 <math>(w_i - w_j)^T x + w_{i0} - w_{j0} = 0</math> 是超平面; 先验会使其平移, 一般不与均值连线正交。</p>                                                             |
| 56        | B         | <p>With <math>\Sigma = \sigma^2 I</math> the boundary is orthogonal to the line linking the two means; with equal priors the optimal rule is the minimum distance classifier, and the perpendicular bisector is <math>x_1 = 2</math>.</p> <p>当 <math>\Sigma = \sigma^2 I</math> 时决策边界与两均值连线正交; 先验相同时最优规则为最小距离分类器, 其垂直平分线为 <math>x_1 = 2</math>。</p>                                                                                                                                                                                                                                 |
| 57        | B         | $\hat{p}(x) = \frac{k}{nV} = \frac{10}{200 \times 0.5} = 0.1; \text{ this follows from } P = \int_R p(t) dt \approx p(x)V \text{ and the estimate } \hat{P} = \frac{k}{n}.$ $\hat{p}(x) = \frac{k}{nV} = \frac{10}{200 \times 0.5} = 0.1; \text{ 它来自 } P = \int_R p(t) dt \approx p(x)V \text{ 以及估计 } \hat{P} = \frac{k}{n}.$                                                                                                                                                                                                                                                       |
| 58        | B         | <p>A sample falls in the hypercube when <math>\left  \frac{x_j - x_{ij}}{h} \right  &lt; \frac{1}{2}</math> for all <math>j</math>, so the kernel is the unit box; the density estimate is then <math>\hat{p}(x) = \left( \frac{1}{nh^d} \right) \sum_i K\left(\frac{x - x_i}{h}\right)</math>.</p> <p>当对任意 <math>j</math> 都有 <math>\left  \frac{x_j - x_{ij}}{h} \right  &lt; \frac{1}{2}</math> 时样本落入超立方体, 因此核为单位方框函数; 密度估计为 <math>\hat{p}(x) = \left( \frac{1}{nh^d} \right) \sum_i K\left(\frac{x - x_i}{h}\right)</math>。</p>                                                  |
| 59        | C         | <p>The kernel must satisfy <math>K(u) \geq 0</math> and the <math>\int K(u) du = 1</math>; therefore any density function can serve as a kernel. The rectangular kernel gives an unsmoothed estimate whereas a Gaussian kernel gives a smooth one.</p> <p>核函数需满足 <math>K(u) \geq 0</math> 且 <math>\int K(u) du = 1</math>, 因此任何概率密度函数都可作为核。矩形核得到不平滑的估计, 高斯核则给出平滑估计。</p>                                                                                                                                                                                                             |
| 60        | E         | <p>Parzen-window estimation selects a region of fixed volume <math>V</math> centred at <math>x</math> and counts <math>k</math>, giving <math>\hat{p} = \frac{k}{nV}</math>; kNN selects a fixed number <math>k</math> and finds the volume <math>V(k)</math> that just encloses them, giving <math>\hat{p} = \frac{k}{nV(k)}</math>.</p> <p>Parzen 窗在 <math>x</math> 处取固定体积 <math>V</math> 的区域并统计样本数 <math>k</math>, 得到 <math>\hat{p} = \frac{k}{nV}</math>; kNN 固定 <math>k</math>, 求出恰好包含 <math>k</math> 个样本的体积 <math>V(k)</math>, 得到 <math>\hat{p} = \frac{k}{nV(k)}</math>。</p> |

| No.<br>题号 | Key<br>答案 | Explanation 解析                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 61        | D         | <p>The width <math>h</math> (bandwidth) sets the spread of each kernel centred at a training sample; the notes show the same samples estimated with different <math>h</math>, which changes the smoothness of the estimated density.</p> <p>宽度 <math>h</math> (带宽) 决定每个以训练样本为中心的核的展开程度; 讲义展示了同一组样本在不同 <math>h</math> 下的估计结果, 密度平滑程度随之改变。</p>                                                                                                                                                                                                                                             |
| 62        | A         | <p>The notes state that the probabilistic neural network and the RBF neural network are almost equivalent to the Parzen-window approach, because both place a kernel function at each training sample.</p> <p>讲义指出概率神经网络与 RBF 神经网络几乎等价于 Parzen 窗方法, 因为二者都在每个训练样本处放置一个核函数。</p>                                                                                                                                                                                                                                                                                                            |
| 63        | E         | <p>The kNN classifier is obtained by substituting the estimates of the prior and the class-conditional density into the Bayes rule, giving <math>\hat{p}(\omega_i x) = \frac{k_i}{k}</math>, which is the majority vote among the <math>k</math> nearest neighbours; for <math>k = 1</math> it is the NN classifier.</p> <p>把先验与类条件密度的估计代入贝叶斯公式得到 <math>\hat{p}(\omega_i x) = \frac{k_i}{k}</math>, 即 <math>k</math> 个最近邻的多数表决; <math>k = 1</math> 时即 NN 分类器。</p>                                                                                                                        |
| 64        | C         | <p>For <math>c = 2</math> the bound is <math>P_c \leq 2P^*(1-P^*) = 2 \times 0.2 \times 0.8 = 0.32</math>; in general <math>P^* \leq P_c \leq P^* \left(2 - \frac{cP^*}{c-1}\right)</math>.</p> <p>当 <math>c = 2</math> 时上界为 <math>P_c \leq 2P^*(1-P^*) = 2 \times 0.2 \times 0.8 = 0.32</math>; 一般形式为 <math>P^* \leq P_c \leq P^* \left(2 - \frac{cP^*}{c-1}\right)</math>。</p>                                                                                                                                                                                                         |
| 65        | C         | <p>For i.i.d. samples the likelihood is <math>p(D \theta) = \prod p(x_k \theta)</math>, and ML selects <math>\hat{\theta} = \operatorname{argmax}_{\theta} p(D \theta)</math>; the logarithm is maximized instead because it is monotonic and turns products into sums.</p> <p>对 i.i.d. 样本, 似然为 <math>p(D \theta) = \prod p(x_k \theta)</math>, ML 取 <math>\hat{\theta} = \operatorname{argmax}_{\theta} p(D \theta)</math>; 由于对数单调且能把乘积变为求和, 通常改为最大化对数似然。</p>                                                                                                                           |
| 66        | D         | <p>Setting the gradient of the log-likelihood to zero gives mean <math>= \left(\frac{1}{n}\right) \sum x_k</math> and variance <math>= \left(\frac{1}{n}\right) \sum (x_k - \text{mean})^2</math>; for the multivariate case <math>\hat{\Sigma} = \left(\frac{1}{n}\right) \sum (x_k - \hat{\mu})(x_k - \hat{\mu})^T</math>.</p> <p>令对数似然的梯度为零, 得到 均值 <math>= \left(\frac{1}{n}\right) \sum x_k</math>, 方差 <math>= \left(\frac{1}{n}\right) \sum (x_k - \text{mean})^2</math>; 多维情形下 <math>\hat{\Sigma} = \left(\frac{1}{n}\right) \sum (x_k - \hat{\mu})(x_k - \hat{\mu})^T</math>。</p> |
| 67        | B         | <p>The prior probabilities are estimated by relative frequencies, <math>\hat{p}(\omega_k) = \frac{n_k}{n} = \frac{30}{200} = 0.15</math>; the class-conditional densities are then estimated separately for each class.</p> <p>先验概率用相对频率估计, <math>\hat{p}(\omega_k) = \frac{n_k}{n} = \frac{30}{200} = 0.15</math>; 随后再对每一类分别估计类条件密度。</p>                                                                                                                                                                                                                                                |

## 匹配题答案 Keys for the matching questions

### 1) Set 1: Image enhancement and denoising filters (Topic 3) 第一组：图像增强与去噪滤波器 (Topic 3)

| Match 配对                                                                       | Matched answer 匹配结果                                                                                                                                                       |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1 -&gt; c)</b><br>Mean filter<br>均值滤波                                      | Attenuates Gaussian noise but blurs edges and spreads impulsive noise<br>能抑制高斯噪声，但会模糊边缘并使脉冲噪声扩散                                                                           |
| <b>P2 -&gt; e)</b><br>Median filter<br>中值滤波                                    | Removes impulsive (salt-and-pepper) noise and preserves step edges, but is weaker than the mean filter for Gaussian noise<br>可去除脉冲（椒盐）噪声并保持阶跃边缘，但对高斯噪声不如均值滤波              |
| <b>P3 -&gt; a)</b><br>Ideal low-pass filter<br>理想低通滤波                          | Sharp cut-off in the frequency domain, which produces ringing artifacts<br>频域截止陡峭，会产生振铃效应                                                                                 |
| <b>P4 -&gt; f)</b><br>Gaussian low-pass filter<br>高斯低通滤波                       | Smooth frequency response $G = \exp \left[ -\frac{u^2+v^2}{2D_0^2} \right]$ used for smoothing<br>频率响应平滑 $G = \exp \left[ -\frac{u^2+v^2}{2D_0^2} \right]$ ，用于平滑          |
| <b>P5 -&gt; d)</b><br>High-boost filter<br>高频提升滤波                              | $f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}$ with $A \geq 1$ , sharpening while keeping low frequencies<br>$f_{hb} = Af - f_{lp} = (A-1)f + f_{hp}$ , $A \geq 1$ , 在保留低频的同时锐化 |
| <b>P6 -&gt; b)</b><br>Iterative truncated mean (ITM) filter<br>迭代截断均值 (ITM) 滤波 | Approaches the median without data sorting, so it is faster than a median filter<br>无需排序即可逼近中值，因此比中值滤波更快                                                                  |

### 2) Set 2: Distances and classifiers (Topics 4 and 5) 第二组：距离与分类器 (Topic 4 与 Topic 5)

| Match 配对                                                                         | Matched answer 匹配结果                                                                                                                         |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1 -&gt; b)</b><br>Squared Euclidean distance<br>平方欧氏距离                       | $d_{Eu} = \ x-\mu\ ^2 = (x-\mu)^T(x-\mu)$<br>$d_{Eu} = \ x-\mu\ ^2 = (x-\mu)^T(x-\mu)$                                                      |
| <b>P2 -&gt; e)</b><br>Squared Mahalanobis distance<br>平方马氏距离                     | $d_{\Sigma} = (x-\mu)^T \Sigma^{-1}(x-\mu)$ , weighted by the inverse covariance<br>$d_{\Sigma} = (x-\mu)^T \Sigma^{-1}(x-\mu)$ ，用协方差矩阵的逆加权 |
| <b>P3 -&gt; d)</b><br>Correlation coefficient of normalized images<br>归一化图像的相关系数 | $\gamma = f_n^T g_n$ , between -1 and 1, equal to 1 for identical patterns<br>$\gamma = f_n^T g_n$ ，取值在 -1 与 1 之间，图案相同时为 1                  |



| Match 配对                                                     | Matched answer 匹配结果                                                                                                                                                                              |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P4 -&gt; f)</b><br>Minimum distance classifier<br>最小距离分类器 | Assigns $x$ to the class whose mean is nearest;<br>equivalent to template matching when priors are<br>equal and $\Sigma = \sigma^2 I$<br>把 $x$ 判给均值最近的类；先验相等且 $\Sigma = \sigma^2 I$ 时等价于<br>模板匹配 |
| <b>P5 -&gt; a)</b><br>Linear classifier<br>线性分类器             | Obtained when all classes share the same<br>covariance matrix; the decision boundary is a<br>hyperplane<br>各类共享同一协方差矩阵时得到，决策边界为超平面                                                               |
| <b>P6 -&gt; c)</b><br>Quadratic classifier<br>二次分类器          | Obtained for Gaussian classes with different<br>covariance matrices; the decision boundary is a<br>quadratic surface<br>各类高斯且协方差矩阵不同时得到，决策边界为二次曲面                                                |

### 3) Set 3: Density estimation and estimation theory (Topic 6) 第三组：密度估计与估计理论 (Topic 6)

| Match 配对                                                       | Matched answer 匹配结果                                                                                                                                    |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1 -&gt; c)</b><br>Parzen-window estimation<br>Parzen 窗估计   | Density estimated by summing kernels centred at<br>the data points, with a fixed bandwidth $h$<br>以数据点为中心叠加核函数来估计密度，带宽 $h$ 固定                          |
| <b>P2 -&gt; f)</b><br>k-nearest-neighbour estimation<br>k 近邻估计 | Density estimated with a neighbourhood whose<br>volume $V(k)$ contains a fixed number $k$ of samples<br>用体积 $V(k)$ 、包含固定数目 $k$ 个样本的邻域来估计密度             |
| <b>P3 -&gt; e)</b><br>Both Parzen and kNN<br>Parzen 与 kNN 的共同点 | Neither assumes a fixed parametric form for $p(x)$ ,<br>so both are nonparametric approaches<br>都不假设 $p(x)$ 具有固定参数形式，因此都是非参数方法                         |
| <b>P4 -&gt; a)</b><br>Maximum likelihood estimation<br>最大似然估计  | Chooses the parameters that maximize the<br>likelihood of the observed training data<br>选择使观测训练数据似然最大的参数                                               |
| <b>P5 -&gt; d)</b><br>Parametric method<br>参数方法                | Assumes an analytical density model with<br>unknown parameters, e.g. a Gaussian with<br>unknown mean and covariance<br>假设具有未知参数的解析密度模型，例如均值与协方差未知的高斯分布 |
| <b>P6 -&gt; b)</b><br>Nonparametric method<br>非参数方法            | No assumption on the form of the density; the<br>estimate is determined entirely by the samples<br>对密度形式不作假设，估计完全由样本决定                                 |